

GOAL: Asset Allocation

Balancing Innovation and Inflation

August 2026

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Tactically neutral, modestly pro-risk for 12m

3-Month Horizon		12-Month Horizon	
Asset Class	Weight*	Asset Class	Weight*
Equities	N	Equities	OW
MSCI Asia Pac ex Japan	↑	MSCI Asia Pac ex Japan	↑
TOPIX	↑	TOPIX	↑
S&P 500	↑	S&P 500	↑
STOXX Europe 600	↓	STOXX Europe 600	↓
10 yr. Gov. Bonds	N	10 yr. Gov. Bonds	N
US	↑	UK	↑
Germany	↑	US	↑
UK	→	Germany	→
Japan	↓	Japan	↓
Commodities	N	Commodities	N
S&P GSCI Precious Metals	→	S&P GSCI Precious Metals	↑
S&P GSCI Energy	→	S&P GSCI Energy	→
S&P GSCI Industrial Metals	→	S&P GSCI Industrial Metals	→
Credit	N	Credit	UW
USD IG	↑	USD IG	↑
USD HY	↑	USD HY	↑
EM USD	→	EM USD	→
EUR IG	↓	EUR HY	↓
EUR HY	↓	EUR IG	↓
Cash	N	Cash	N

* Arrows denote preferences within asset classes.

Source: Goldman Sachs Global Investment Research

Global growth/inflation has deteriorated due to the Middle East war but should improve with inflation normalising

Real GDP growth

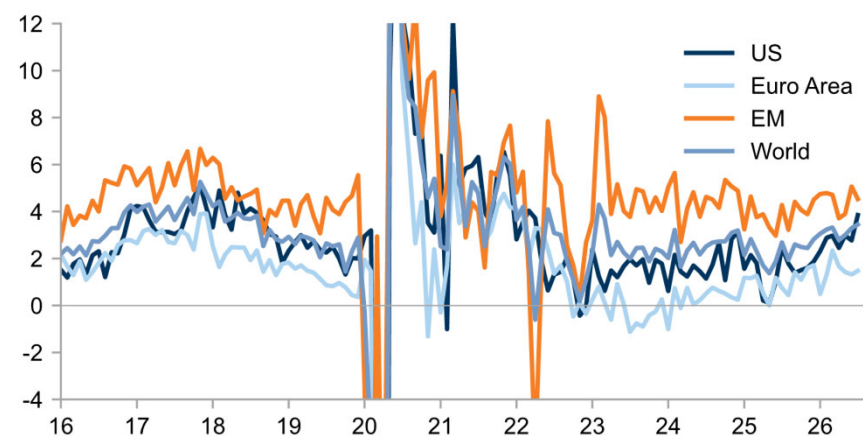
% yoy	2025	2026E		2027E	
	GS	GS	Cons.*	GS	Cons.*
USA	2.1	2.1	2.2	2.3	2.1
Japan	1.1	0.5	0.6	1.1	0.8
Euro area	1.3	0.8	0.7	1.2	1.2
UK	1.3	1.1	1.0	1.3	1.2
China	5.0	4.6	4.6	4.7	4.4
India	7.5	6.9	-	7.4	-
Brazil	2.3	2.2	2.0	1.4	1.8
Russia	1.0	0.2	0.7	1.3	1.3
Advanced Economies	1.8	1.6	1.6	1.8	1.8
Emerging Markets	4.2	3.7	3.7	4.1	4.3
World	2.8	2.5	2.9	2.8	3.1

*Bloomberg country and GS-weighted aggregate consensus

Headline Inflation

CPI % yoy	2024	2025E	2026E	2027E
USA	3.0	2.7	3.2	2.3
Japan	2.7	3.2	1.8	1.4
Euro area	2.4	2.1	2.9	2.4
UK	2.5	3.4	3.0	2.2
China	0.2	0.1	1.0	1.0
India	5.0	2.2	4.3	4.7
Brazil	4.4	5.0	4.5	4.6
Russia	8.5	8.7	6.0	5.2
Advanced Economies	2.7	2.5	2.9	2.2
Emerging Markets	5.9	4.0	4.3	3.8
World	4.6	3.4	3.7	3.2

Current Activity Indicator (CAI)



Financial Condition Indices (FCI)

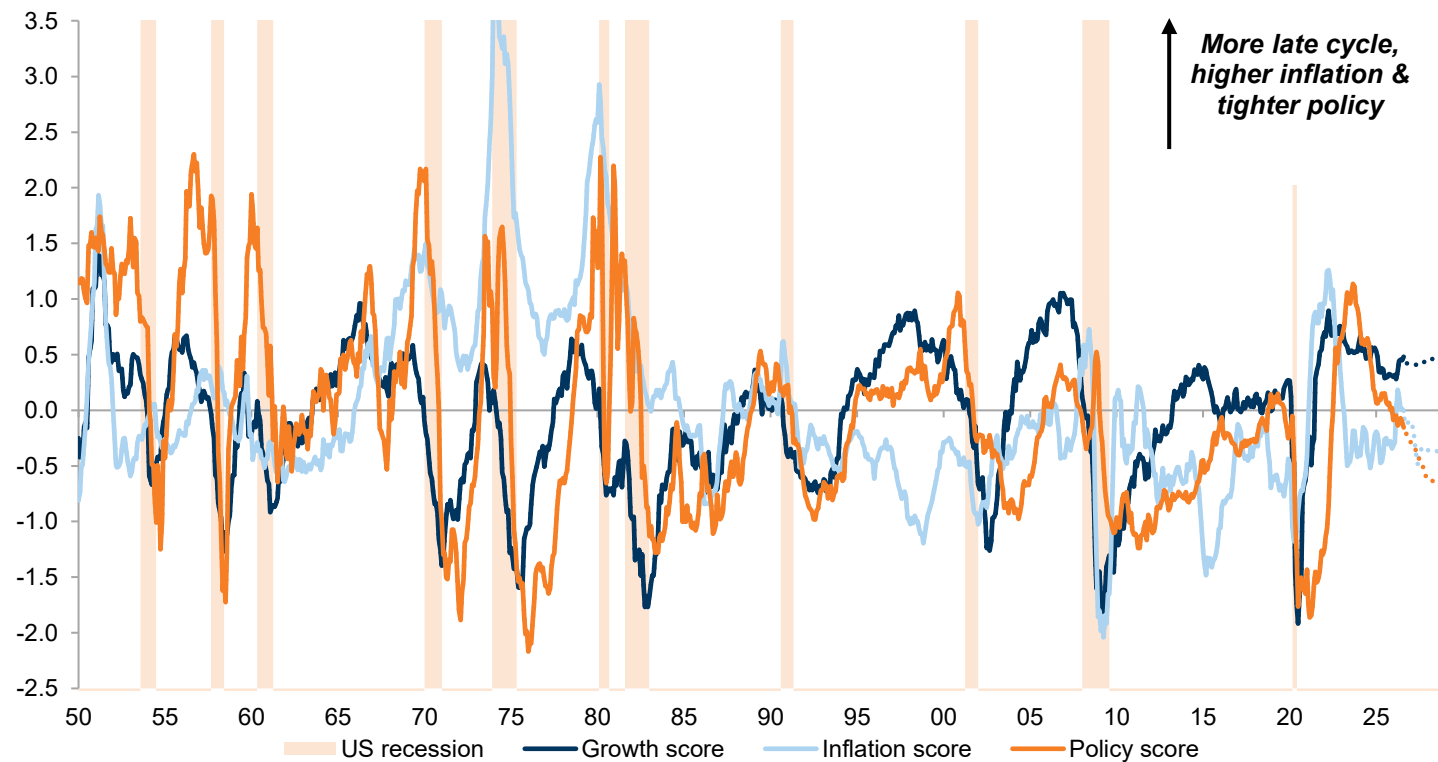


Source: Bloomberg, Goldman Sachs Global Investment Research

Business cycle shifting from Goldilocks to Reflation...and back

Growth, inflation and policy can disconnect along the business cycle

Average expanding z-score of macro and market variables across growth, inflation, and policy

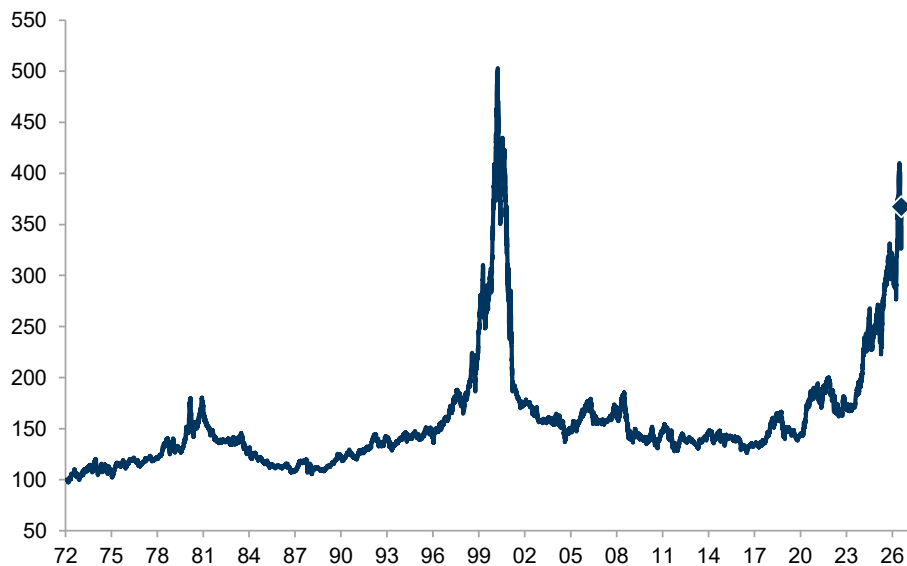


Source: Haver Analytics, Bloomberg, Datastream, Goldman Sachs Global Investment Research

The Momentum rally has been nearing Tech Bubble levels but volatility has picked up into the summer

The Momentum stock rally has been nearing Tech Bubble levels...

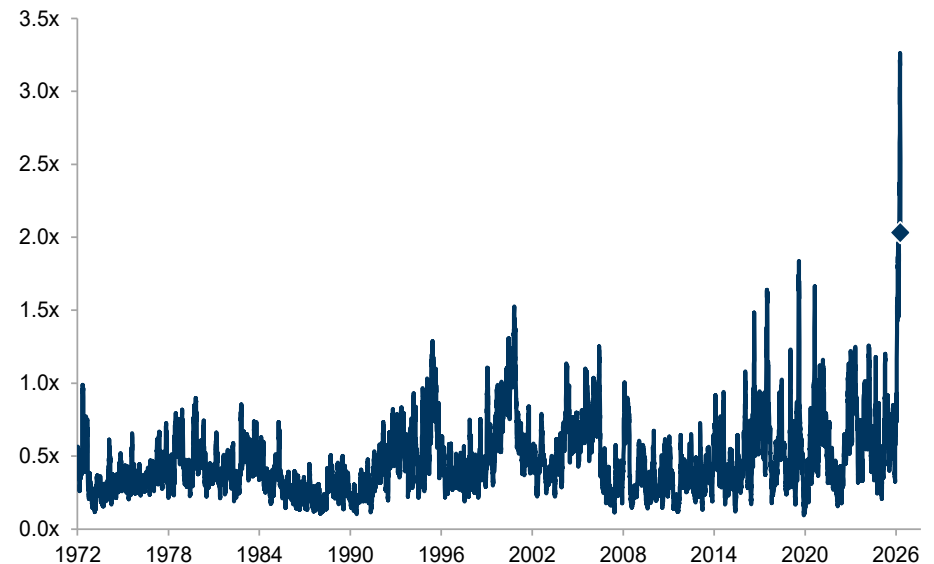
Relative total return performance S&P 500 momentum vs. low volatility stocks



...but volatility has picked up across stocks, despite low market volatility

1-month realised vol of the momentum factor / 1-month realised vol of S&P 500.

Momentum factor = S&P 500 Momentum vs. market

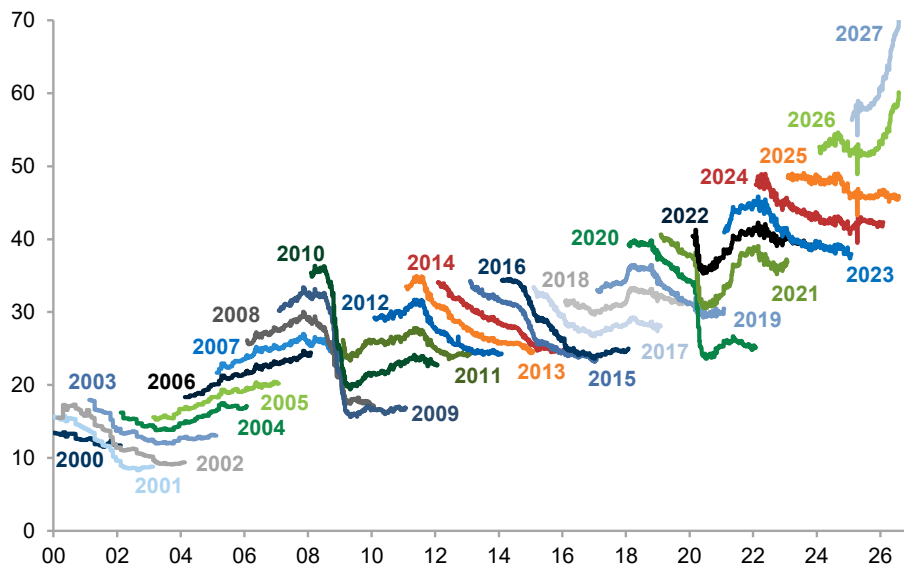


Source: Datastream, Bloomberg, Goldman Sachs Global Investment Research

More than during the Tech Bubble earnings have been the key driver for equity returns

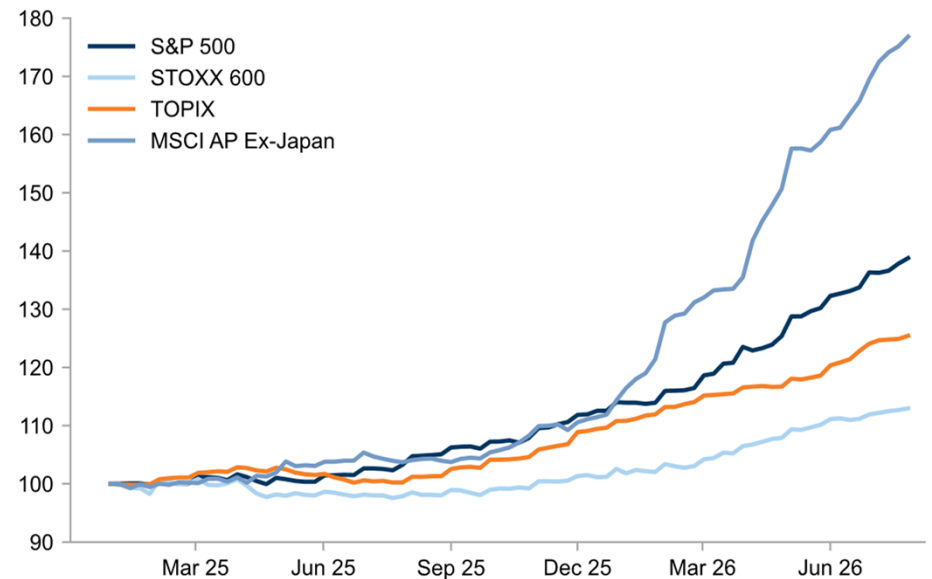
Global earnings revisions have been unusually positive YTD

Consensus estimates in USD. MSCI AC World EPS



Earnings have been revised higher, led by Asia ex Japan and the US

Consensus 12-month forward EPS. Indexed to 100 in January 2025

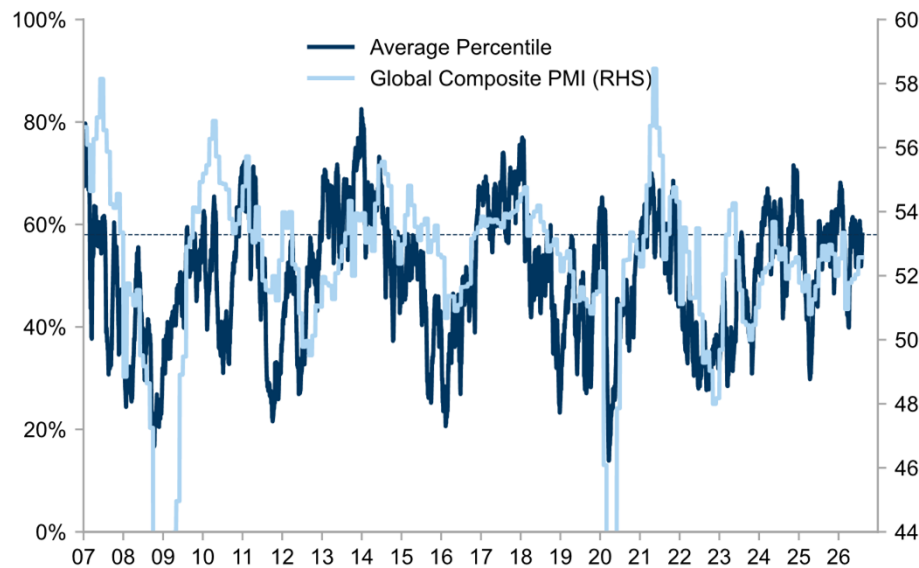


Source: Datastream, Bloomberg, Goldman Sachs Global Investment Research

Cross-asset sentiment and positioning are back to more neutral levels

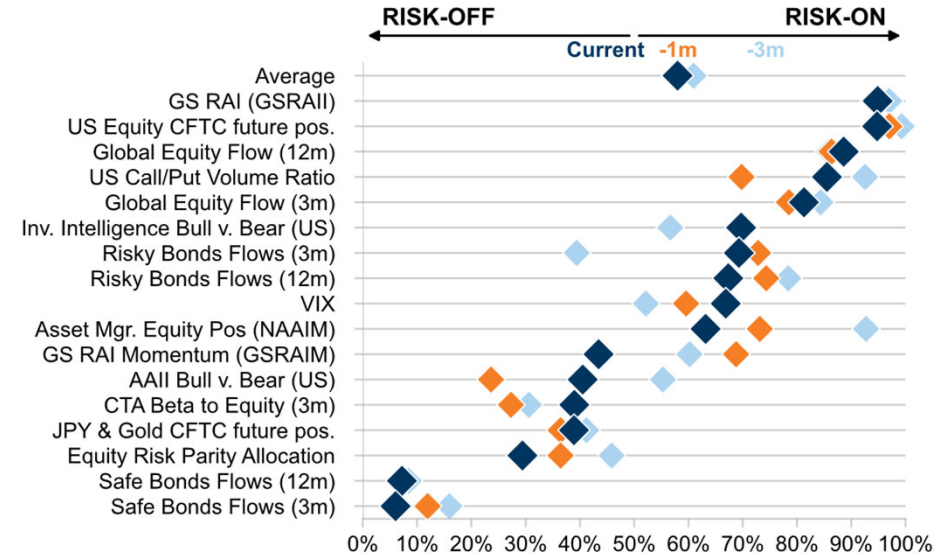
Average percentile of sentiment indicators

Data since 2007



Cross-asset positioning and sentiment indicators

Percentile of sentiment indicators since 2007

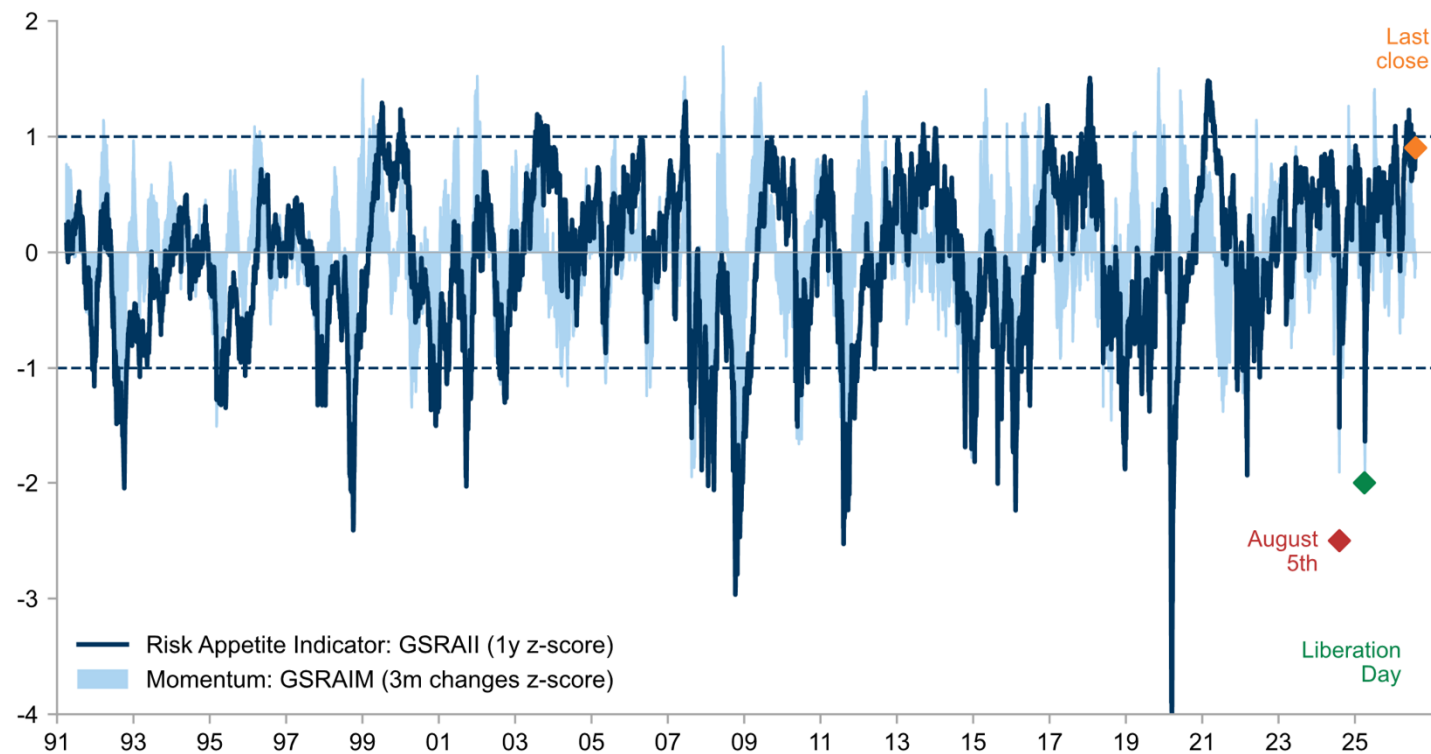


Source: Haver Analytics, EPFR, Datastream, Goldman Sachs Global Investment Research

Our Risk Appetite Indicator remains at elevated levels

Our Risk Appetite Indicator rose above 1.2 at the start of June

GS Risk Appetite Indicator (GSRAII Index)

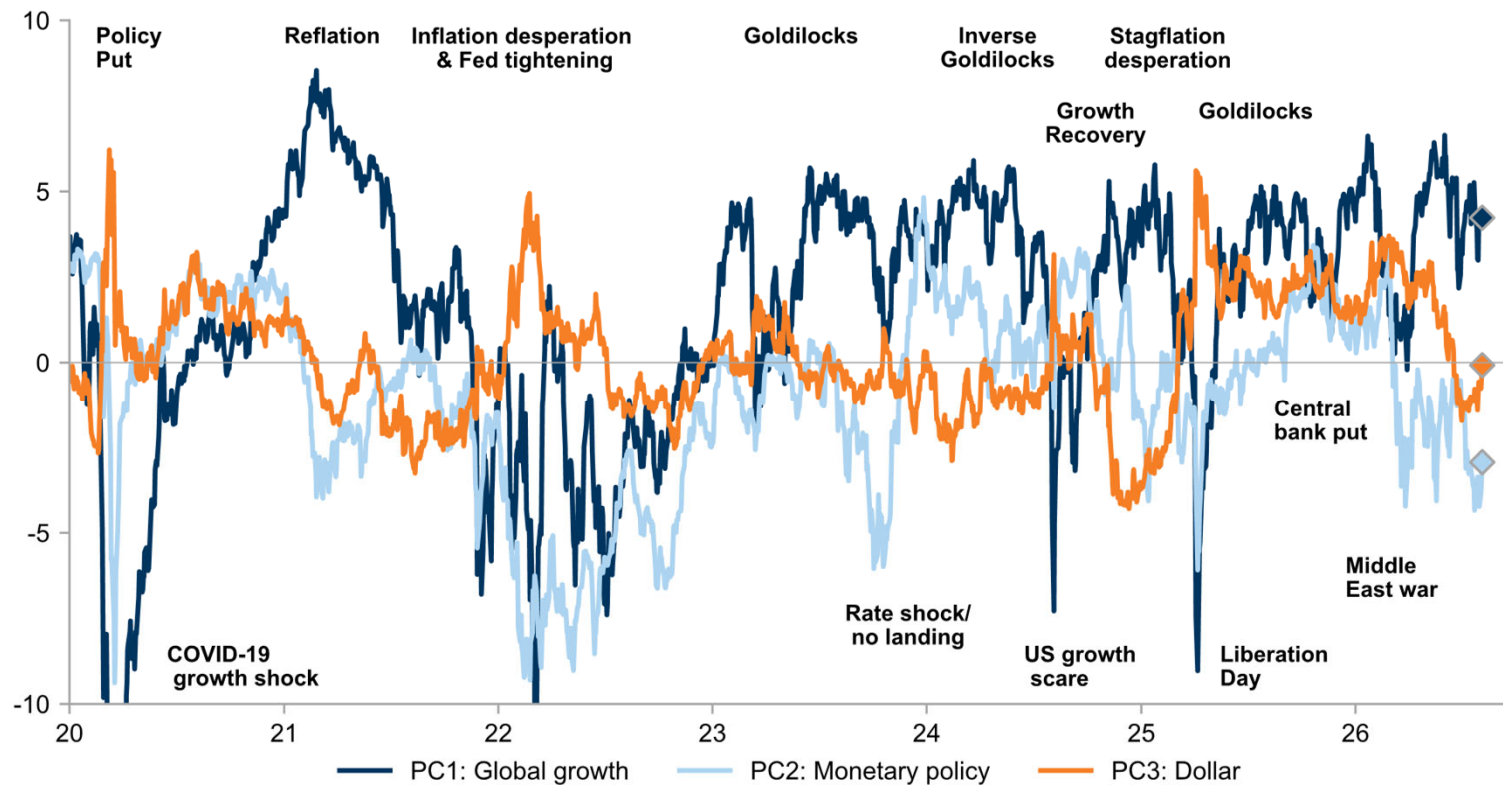


Source: Haver Analytics, Datastream, Goldman Sachs Global Investment Research

Markets are pricing a more 'reflationary' backdrop with growth optimism but less dovish monetary policy

Market narrative volatility has been high with large shifts in cross-asset pricing

RAI principal components

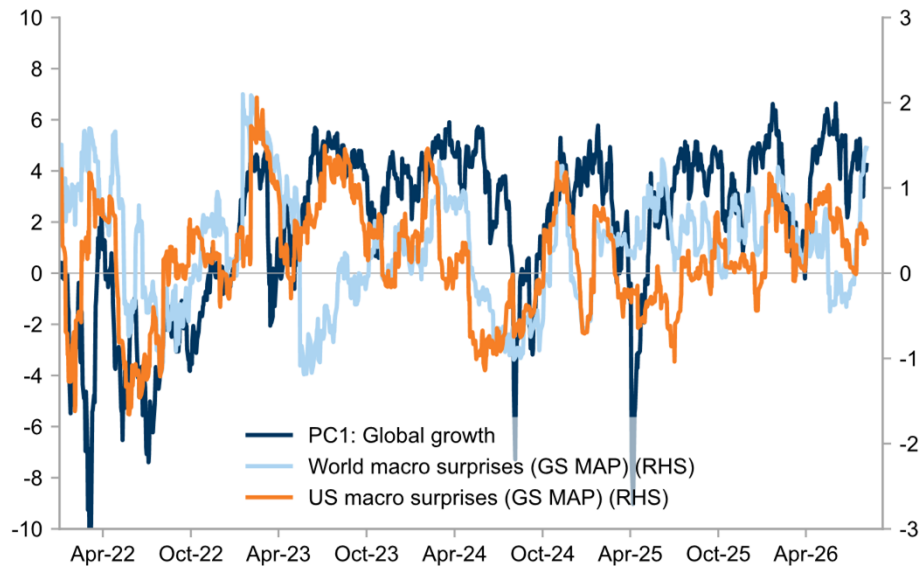


Source: Goldman Sachs Global Investment Research

Growth optimism has been ahead of macro surprises, rates relief closely linked to central bank pricing

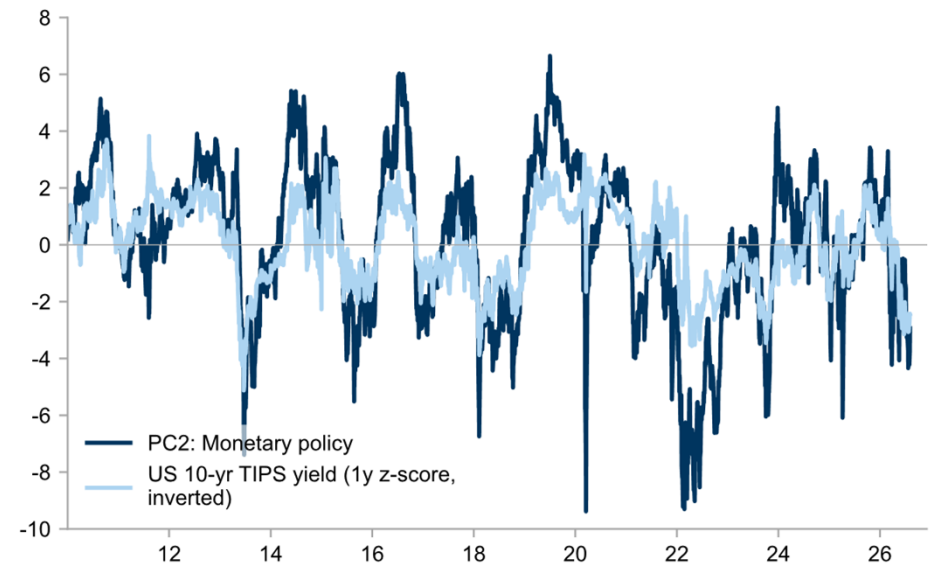
Growth pricing has picked up despite before macro surprises

PC1: Global growth factor vs. Global and US MAP Scores



Rising long-dated real yields (all else equal) weigh on risk appetite

PC2: Monetary policy factor vs. 1-year rolling z-score of US 10-year TIPS yield

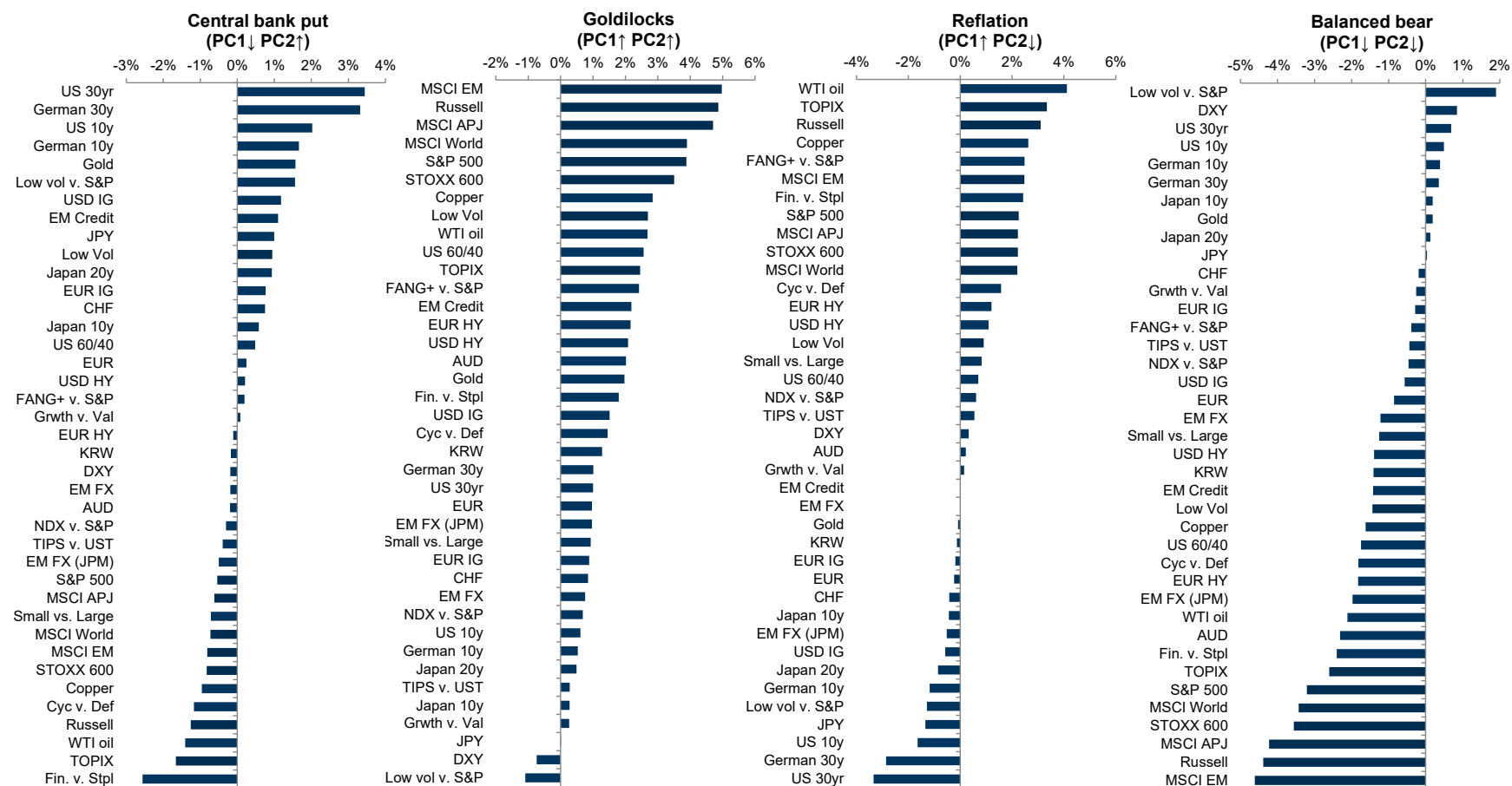


Source: Datastream, Goldman Sachs Global Investment Research

Markets priced a “Balanced Bear” regime in March and shifted towards ‘Reflation’ since then

Changing drivers of risk appetite could drive a material rotation across and within assets

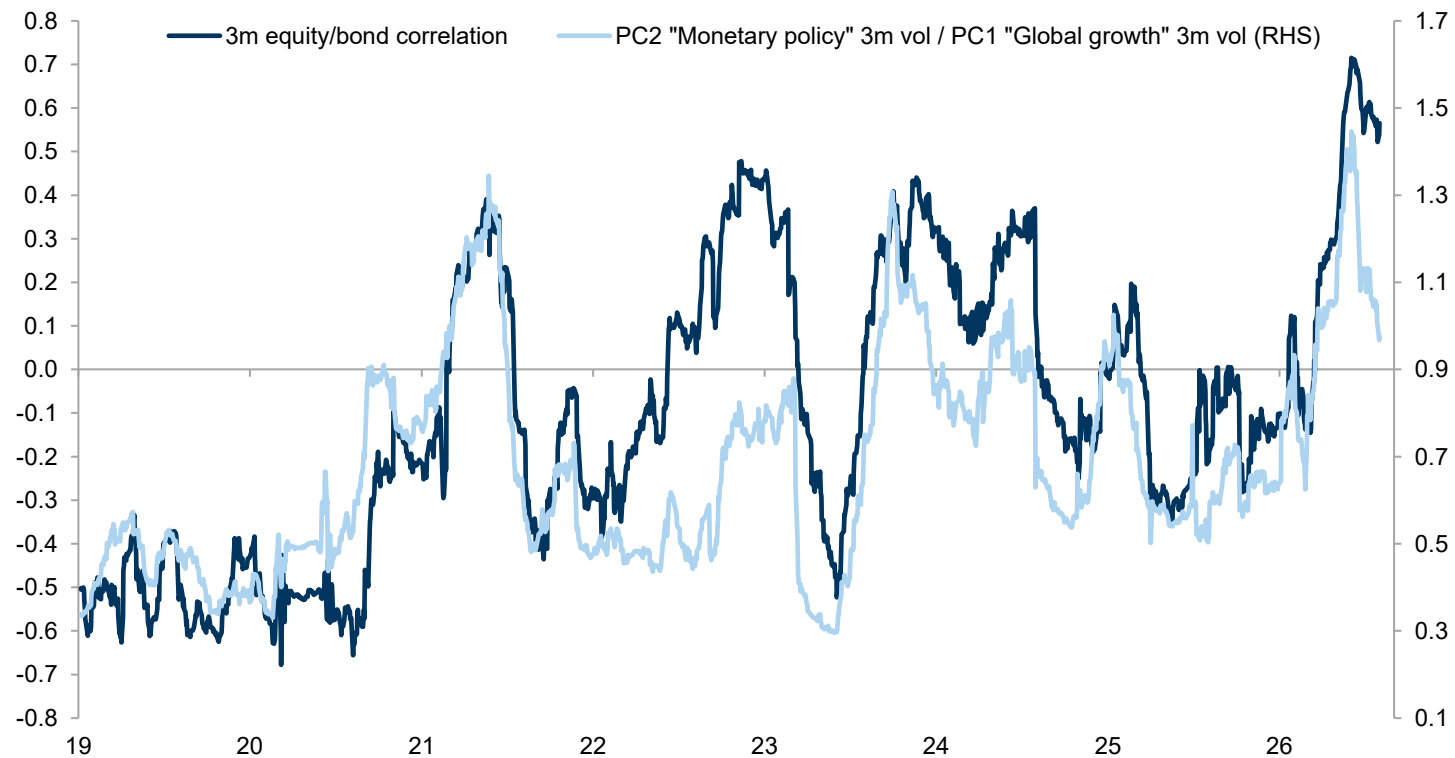
Average monthly returns for different RAI PC1 "Global Growth" and PC2 "Monetary Policy" mixes since 2000



Source: Datastream, Bloomberg, Haver Analytics, Goldman Sachs Global Investment Research

Equity/bond correlations turned positive as markets priced more of a rate rather than a growth shock during the Middle East war

Equity/bond correlations more positive with rate shocks becoming more important vs. growth shocks

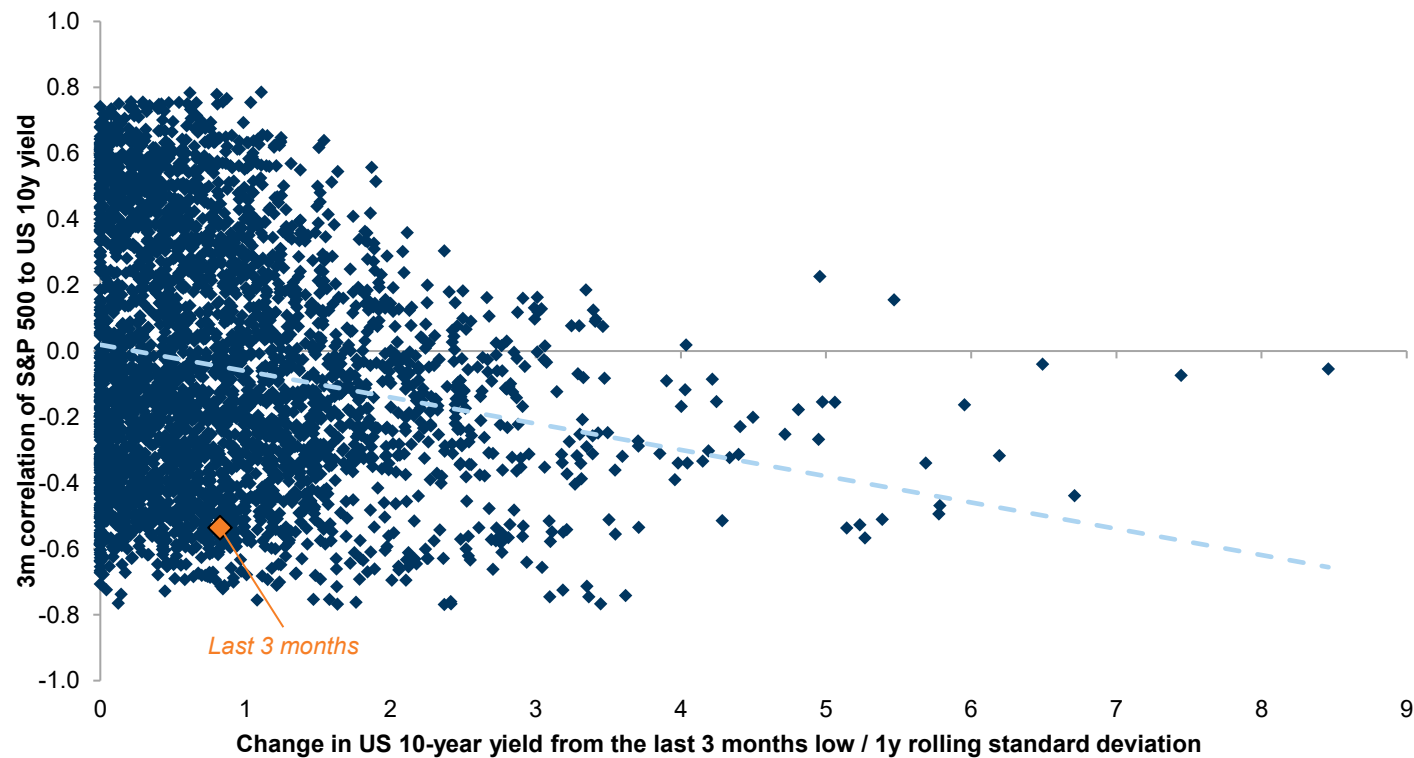


Source: Datastream, Haver Analytics, Goldman Sachs Global Investment Research

The speed of bond yield increases matters

US 10-year yield increases above 2 standard deviations tend to weigh on equities

Data since 1962

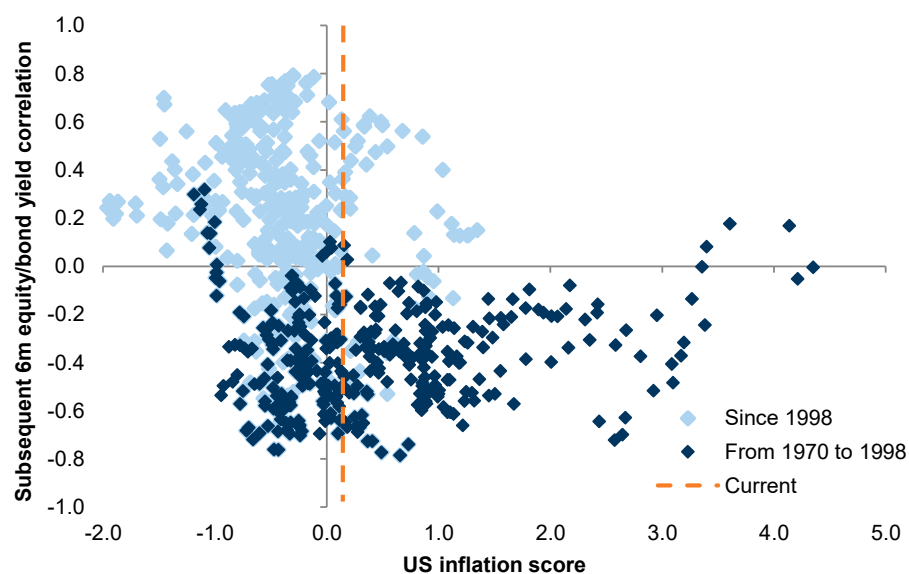


Source: Datastream, Goldman Sachs Global Investment Research

Rising bond yields due to sticky inflation and fiscal concerns are difficult to digest for equities

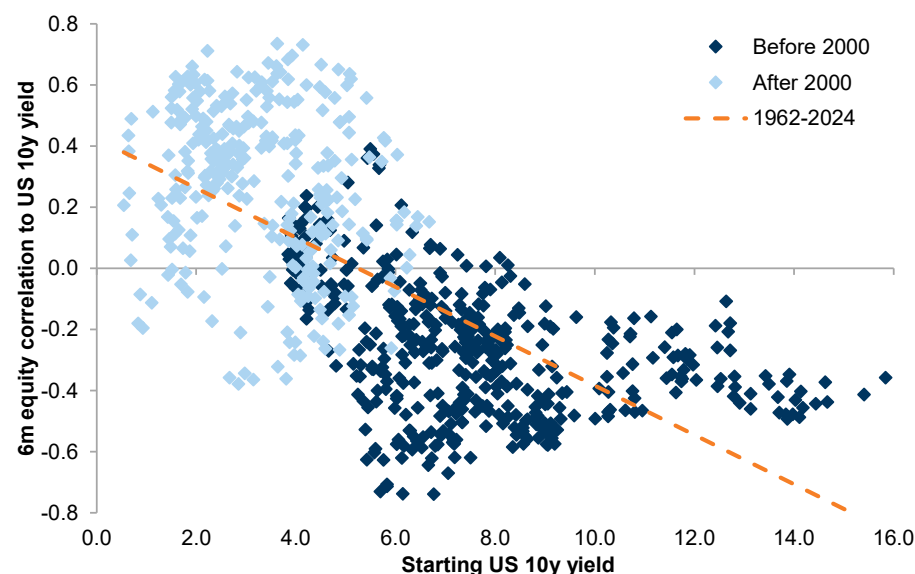
If the inflation score is positive, the equity/bond yield correlation tends to be more negative

Equity/bond correlation and inflation score



Above 5% US 10-year yields equity/bond yield correlations have been more negative

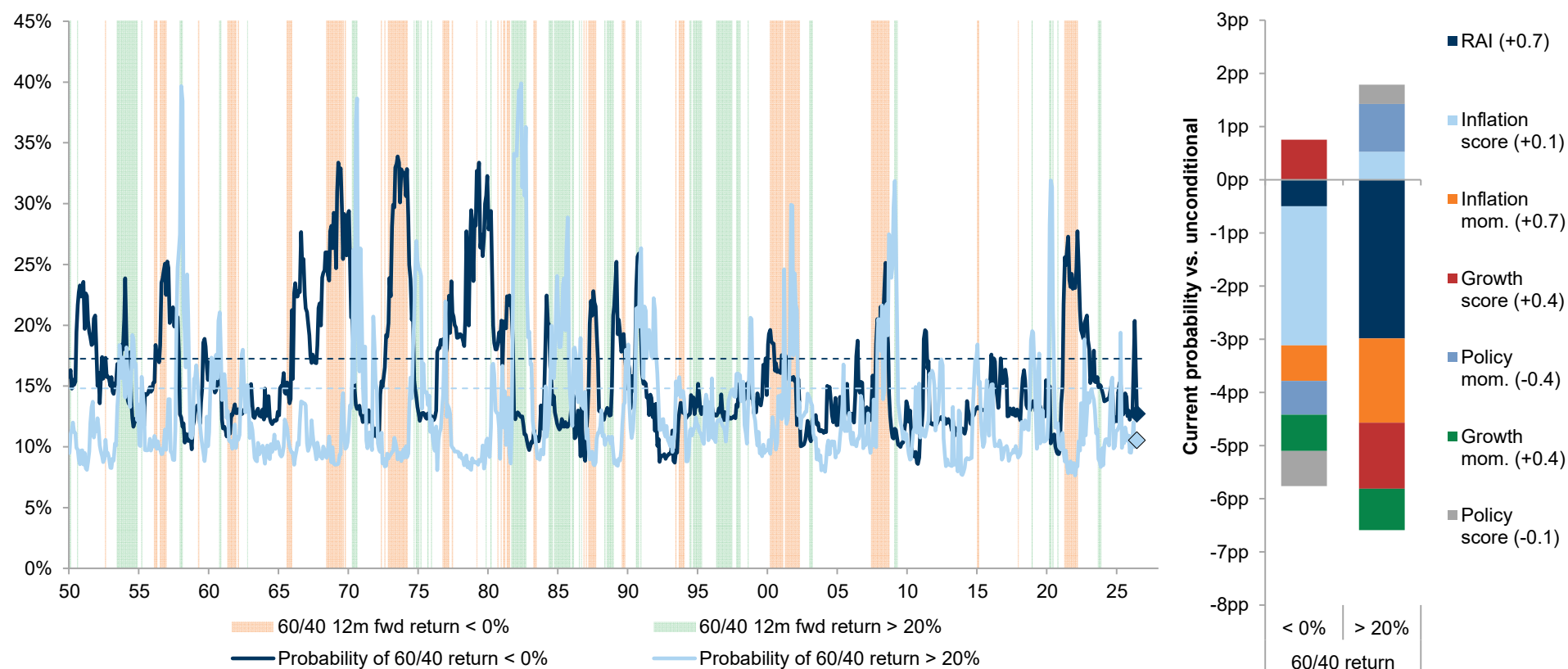
Data since 1962



Source: Bloomberg, Haver Analytics, Goldman Sachs Global Investment Research

Risk of a stagflationary shock and downside risks for 60/40 portfolios have moderated

Based on our dynamic asset allocation model, the risk of 60/40 portfolio drawdown did pick up in function of rising inflation
Random forest probability (dashed lines = unconditional probabilities); bar chart shows the contribution of each cycle score to the current probability vs. its unconditional level (latest score in brackets)

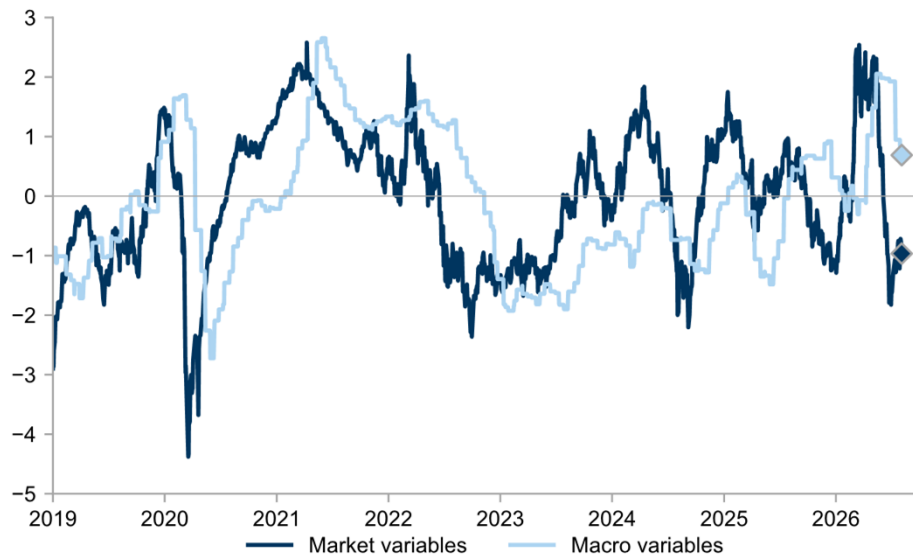


Source: Haver Analytics, Bloomberg, Goldman Sachs Global Investment Research

Inflation momentum should turn more negative in the second half of the year

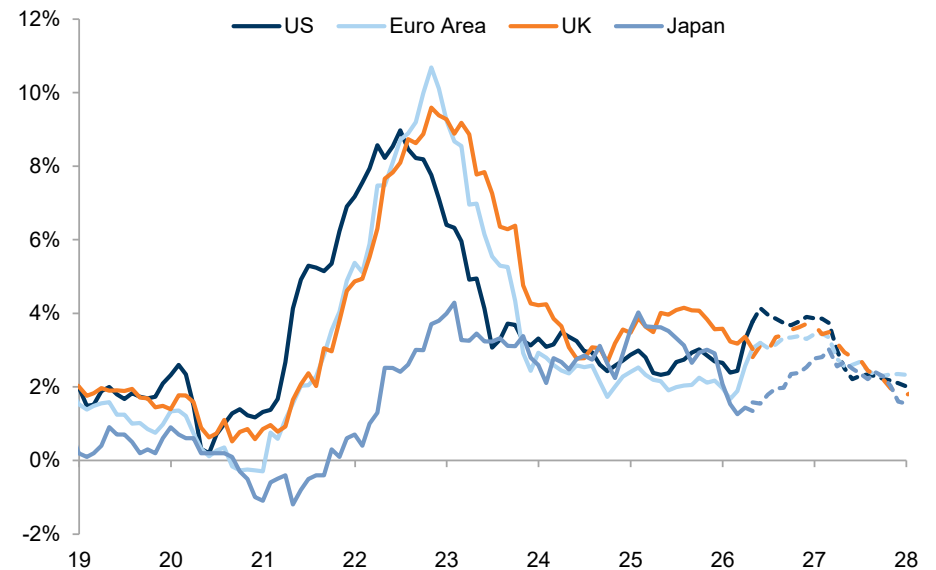
Market measures of inflation have started to normalise but macro data not yet

Inflation momentum: 1-year rolling z-score of measures of US inflation



Our economists forecast headline inflation staying elevated until year-end before normalising

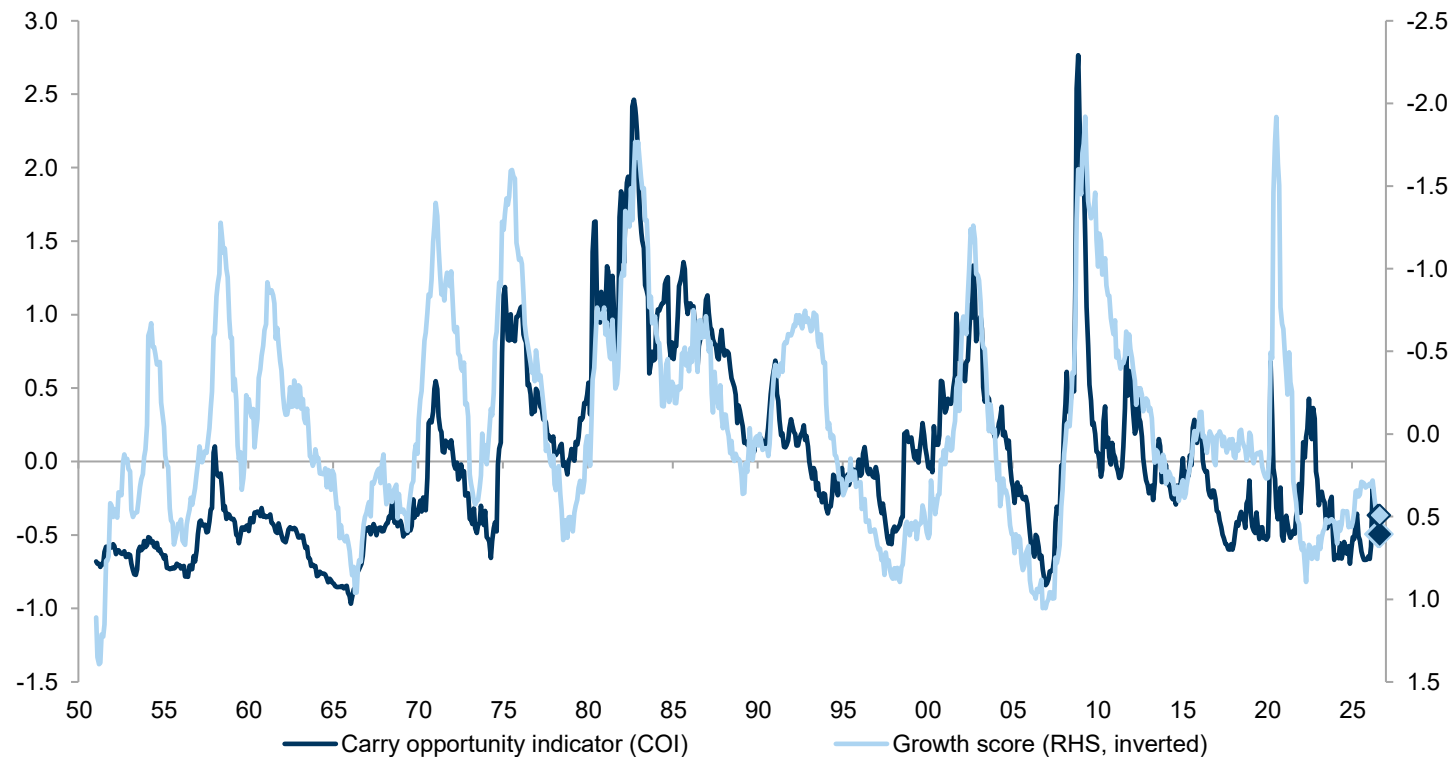
CPI yoy inflation (dotted lines denote our economists' forecasts)



Source: Haver Analytics, Bloomberg, Goldman Sachs Global Investment Research

Current low risk premia signal poor asymmetry for carry trades and increased vulnerability to both growth and rate shocks

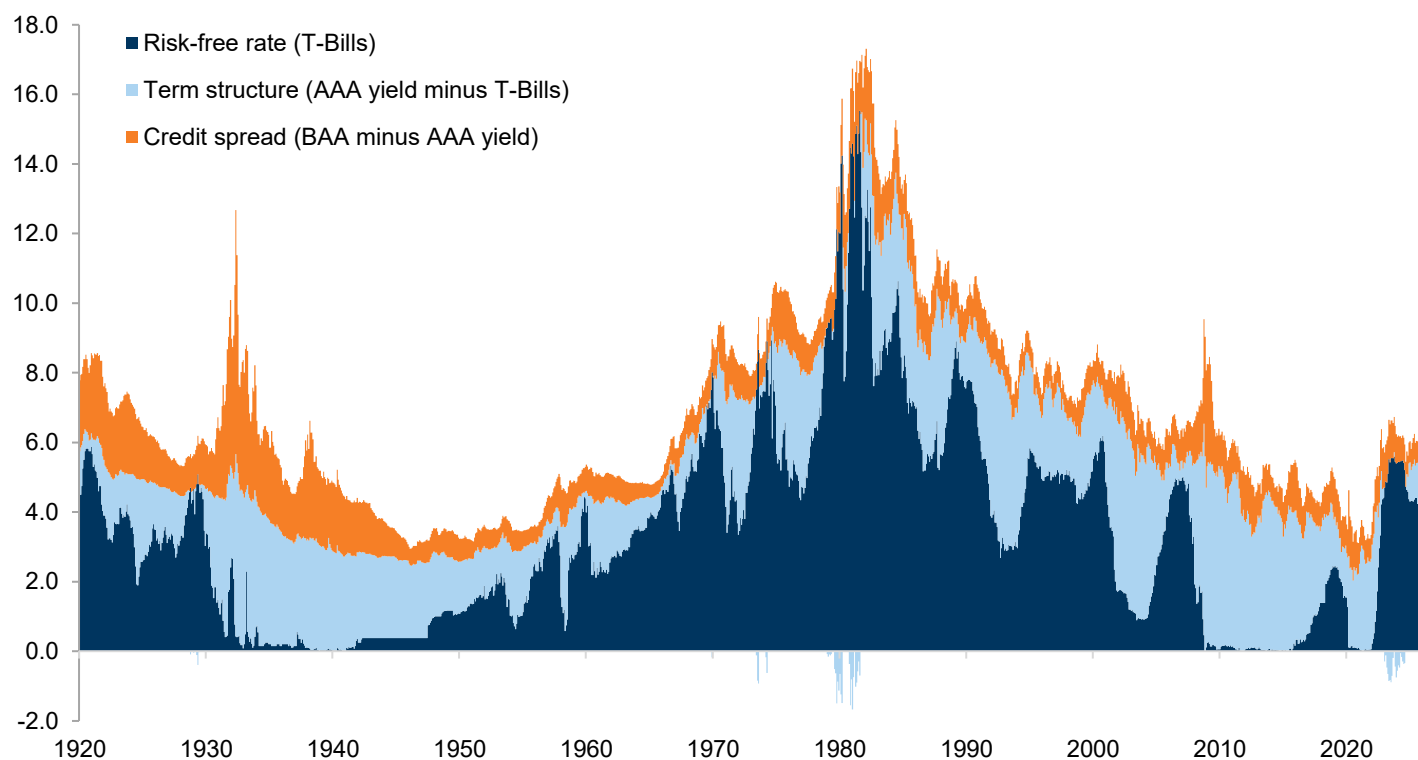
Our Carry Opportunity Indicator (COI), which includes EM/sovereign/corporate credit, FX, equity volatility and yield curve, is near its lowest levels since the GFC



Source: Datastream, Haver Analytics, Goldman Sachs Global Investment Research

All-in credit yields are elevated, but credit spreads and term premia are relatively low

Risk-free rates, term structure, and credit spreads

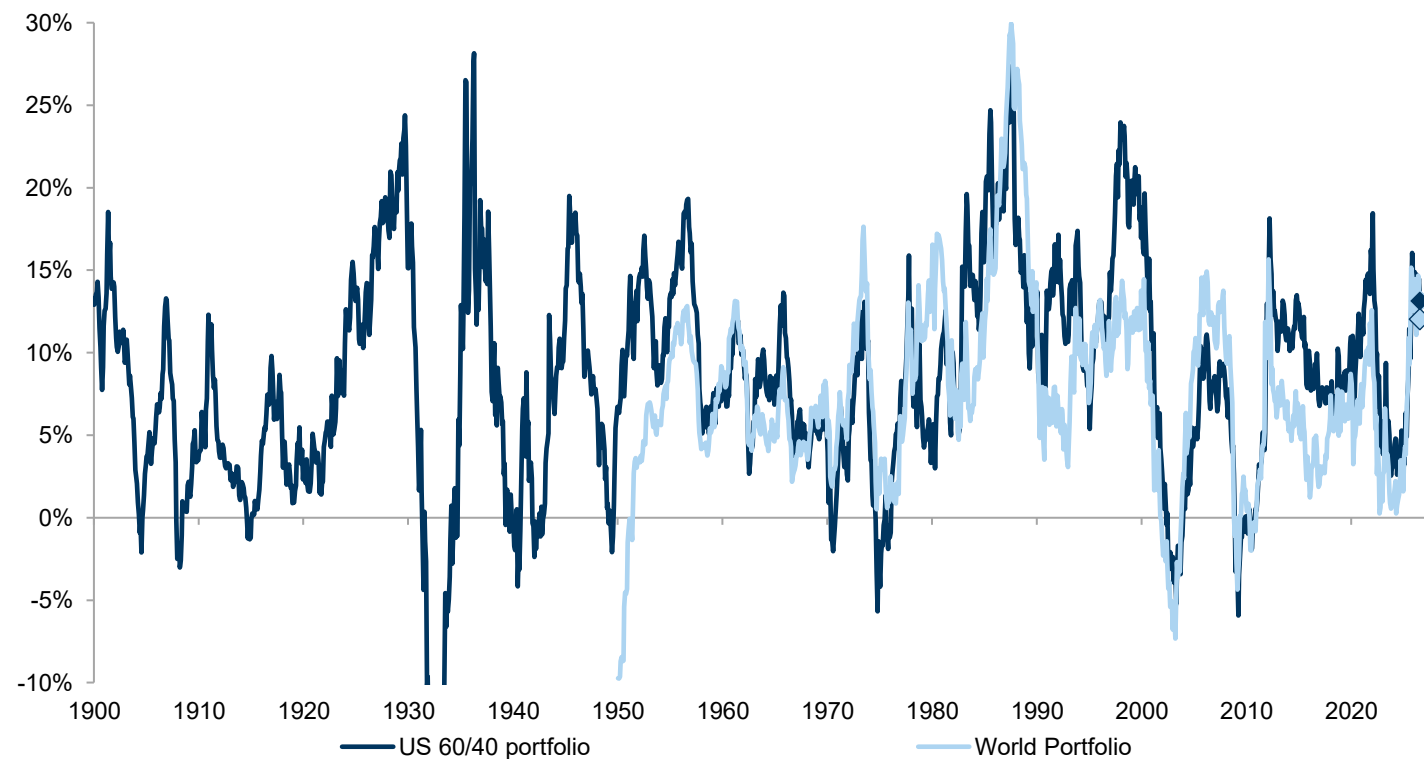


Source: Datastream, Bloomberg, Haver Analytics, Goldman Sachs Global Investment Research

Balancing Innovation and Inflation in Portfolios

Balanced portfolios have delivered strong returns in recent years

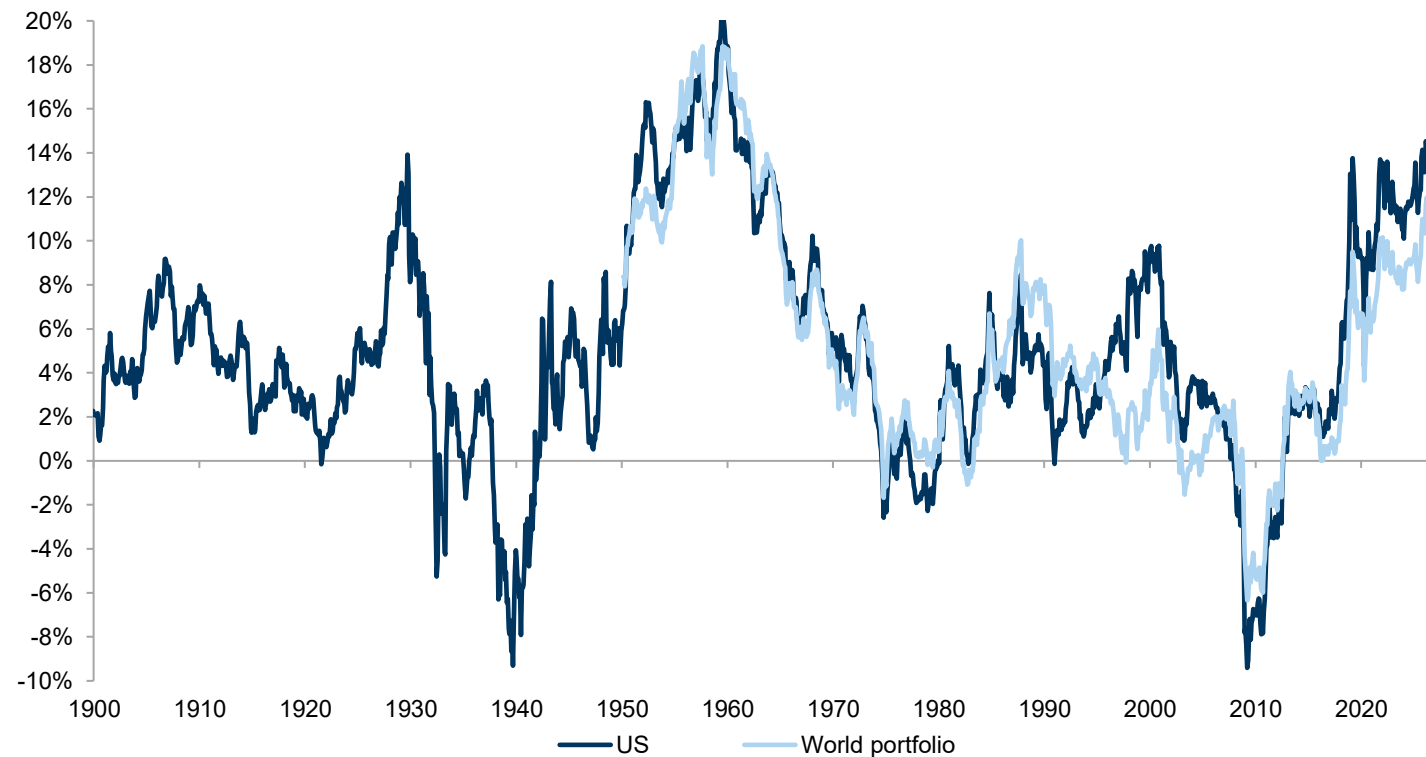
3-year rolling annualised nominal return (US\$)



Source: Datastream, Haver Analytics, Goldman Sachs Global Investment Research

Equity risk premia have been among the strongest since the Roaring 1920s and 1950s

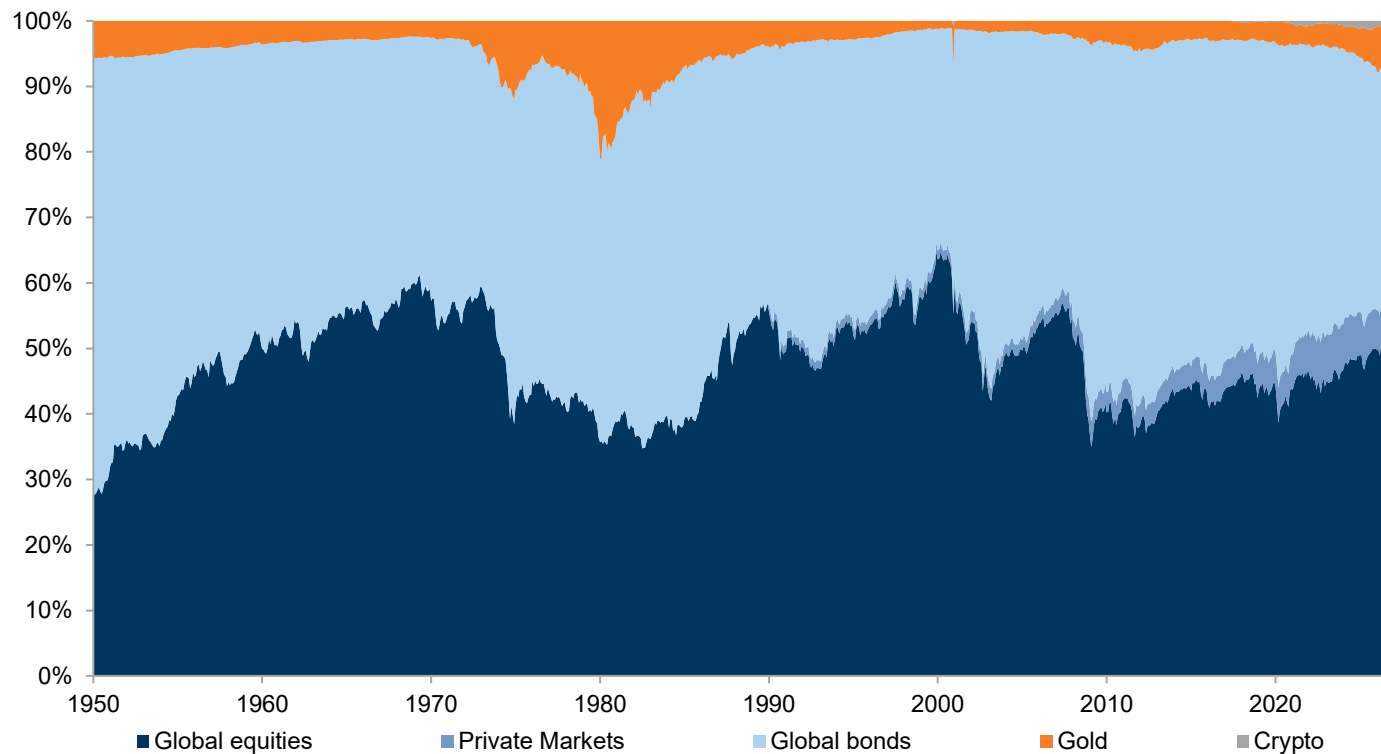
10-year rolling annualised equity vs. bond performance (ex post equity risk premia)



Source: Robert Shiller, Datastream, Haver Analytics, Goldman Sachs Global Investment Research

The World Portfolio has shifted to equities, especially when including private markets

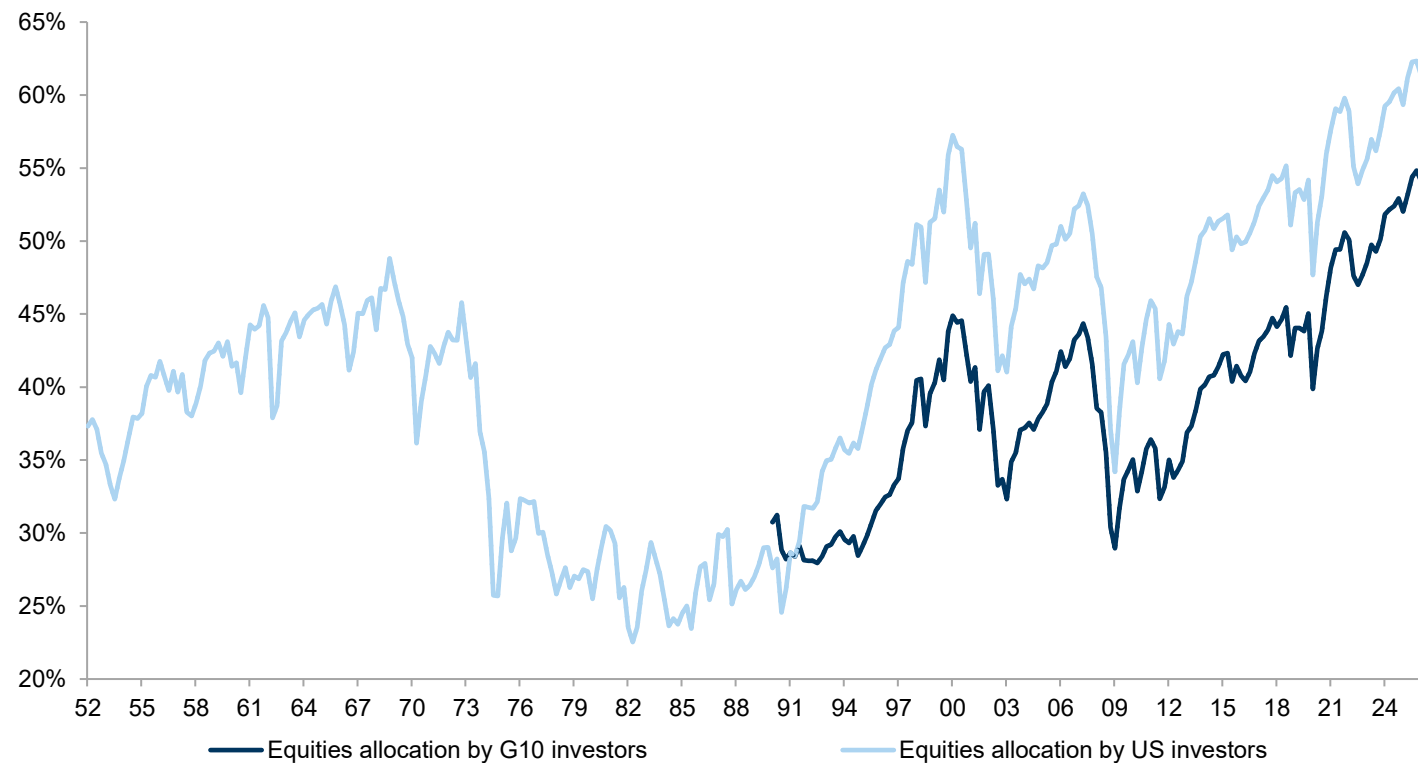
Relative asset weights in the World Portfolio (current coverage US\$289trn assets)



Source: Bloomberg, Datastream, Haver Analytics, Preqin, Goldman Sachs Global Investment Research

Investor allocations to equities have picked up in line with shifts in the World Portfolio

Aggregate G10/US investors' equity allocation (households, pension and insurance and investment funds)

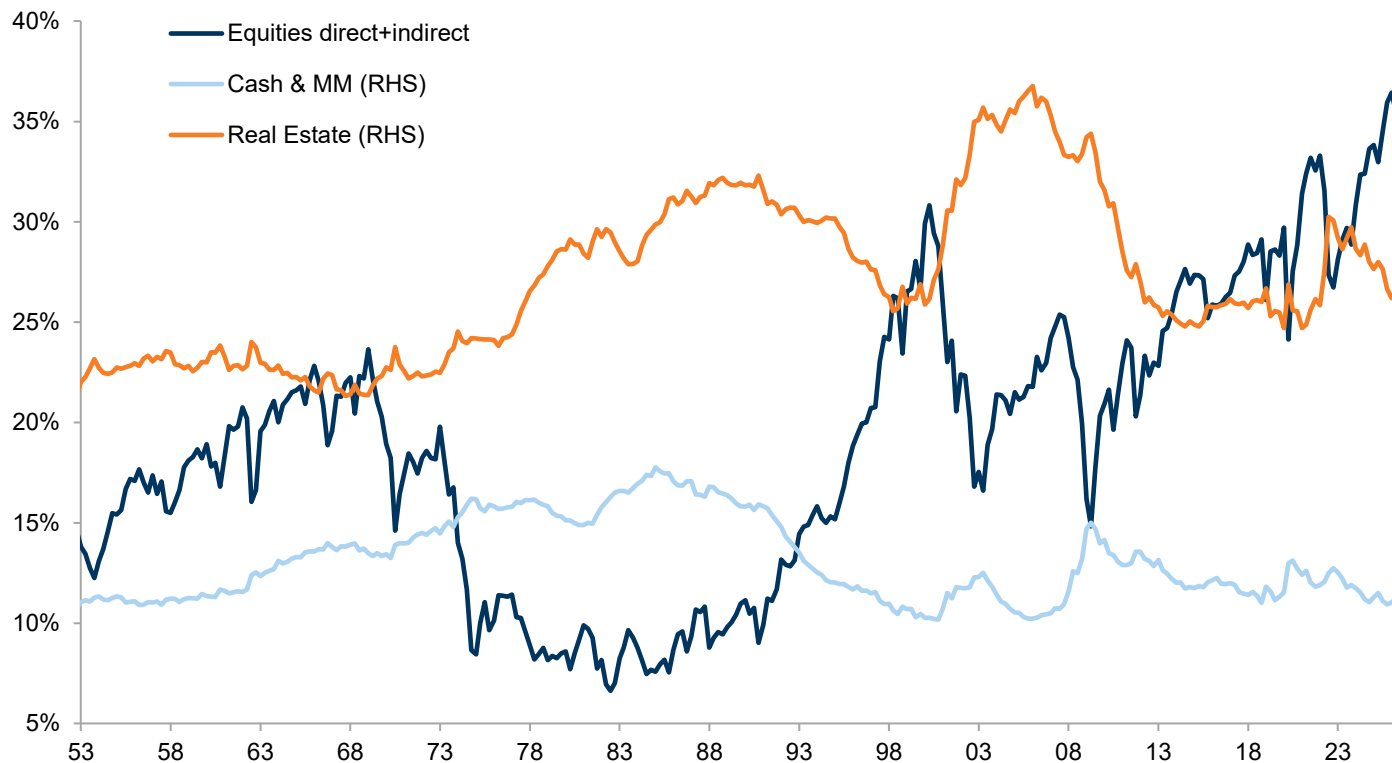


Note: Split made on equity, bond and cash allocation from HH/NPISH, non-MMF investment funds and insurance/pensions.

Source: Haver Analytics, Goldman Sachs Global Investment Research

Equity holdings have overtaken real estate as proportion of household net financial wealth

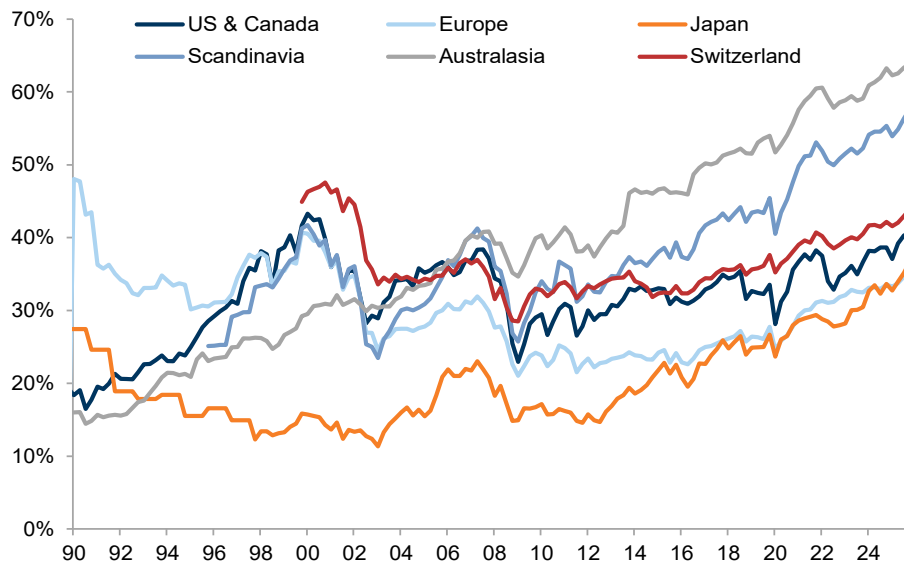
Quarterly data, as of Q1 2026, expressed as % of household net financial wealth



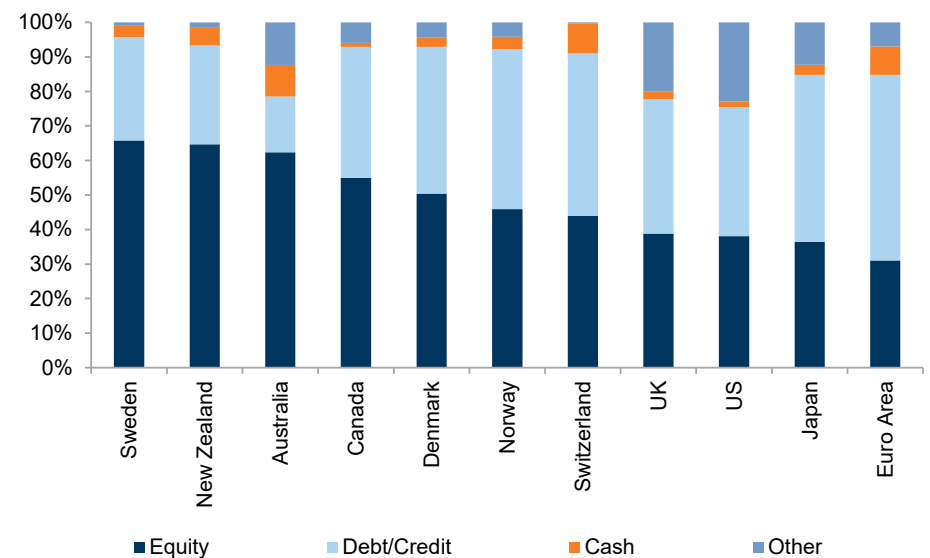
Source: Haver Analytics, Goldman Sachs Global Investment Research

Large divergences in pensions and insurance allocations – regulatory shifts for more equity allocations in Europe

Large divergences in equity allocations among pensions and insurance
Equity allocation as % of total assets for insurance and pensions



Weights as % of insurance and pensions' total assets

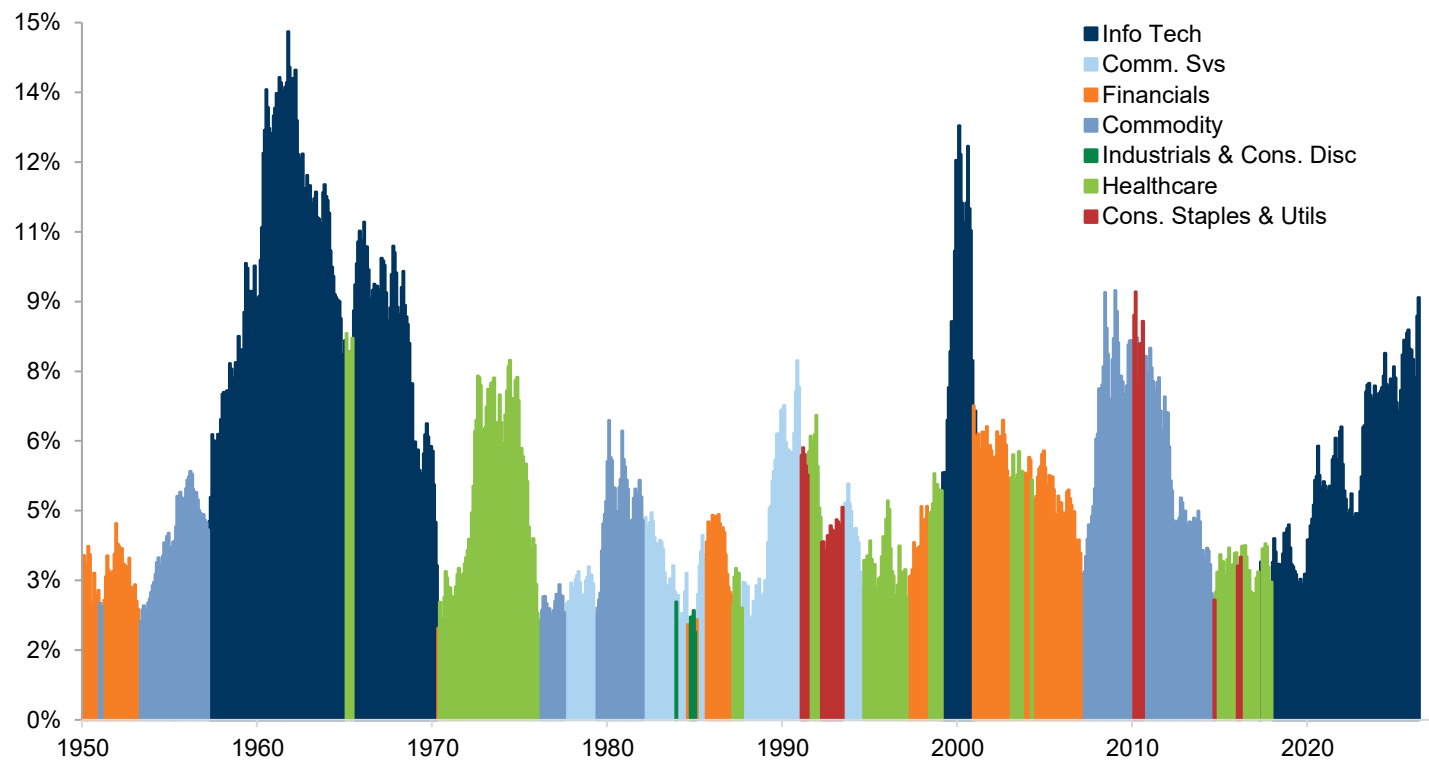


Note: Allocation projected to date with benchmark performance from Q1 2026.

Source: Haver Analytics, Goldman Sachs Global Investment Research

Tech sector outperformance has been unusually strong and prolonged

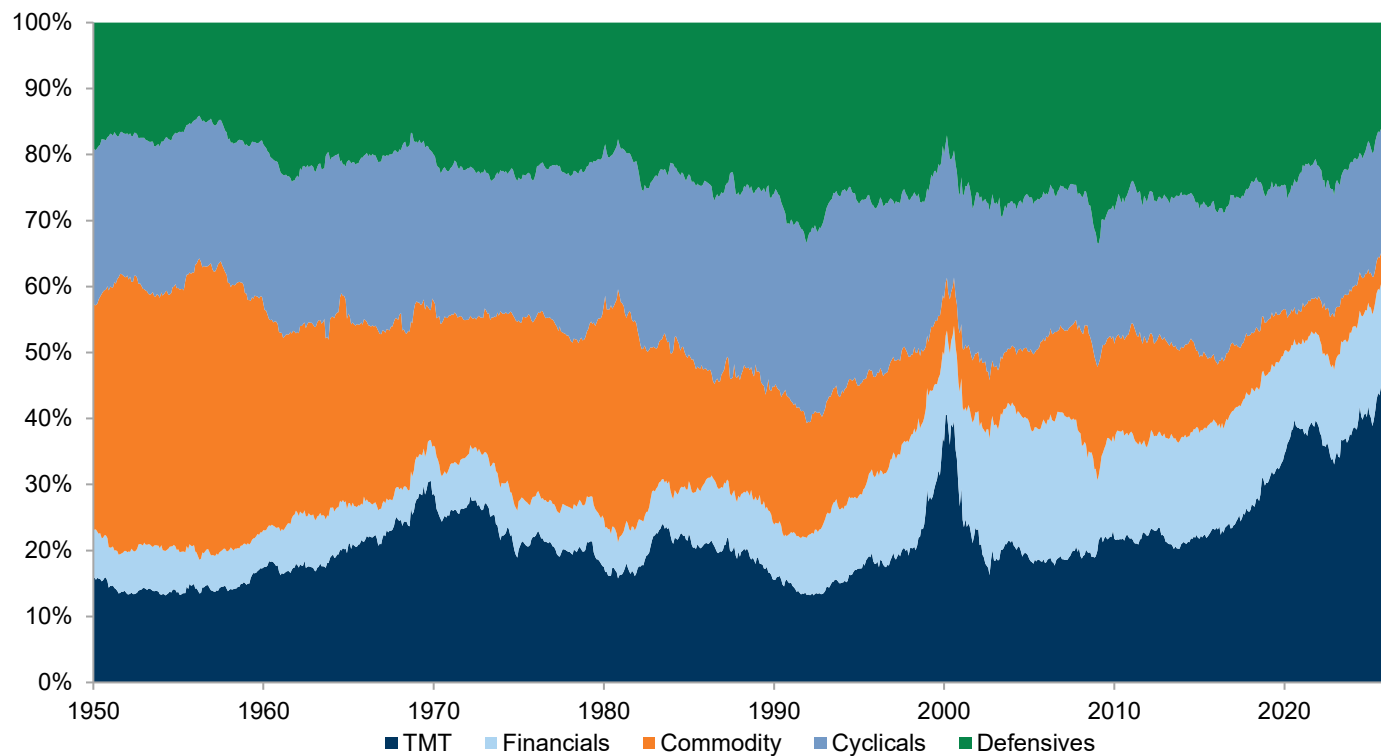
10-year annualised performance vs. the market of the best performing US sector



Source: Kenneth French, Goldman Sachs Global Investment Research

The Tech sector equity weight is now above Tech Bubble levels

Relative sector weights within US equities

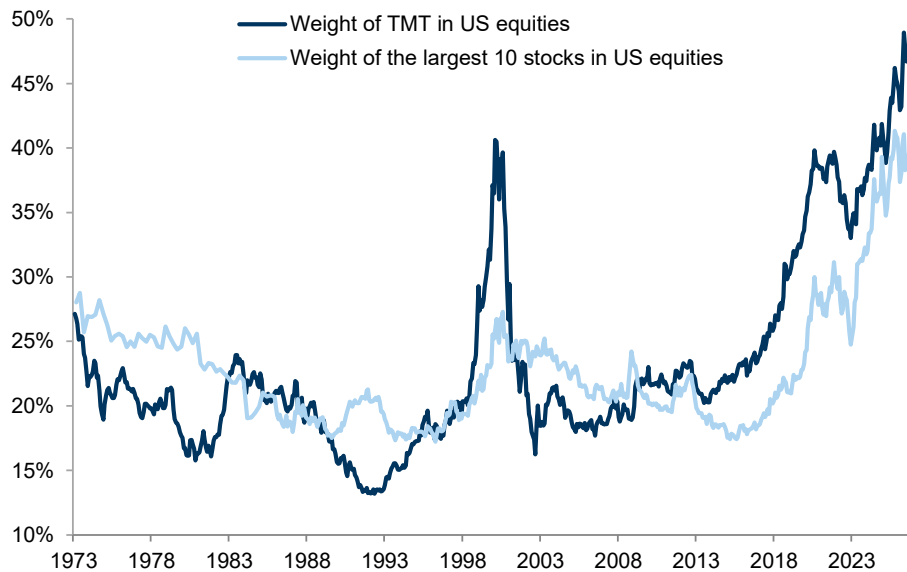


Source: Bloomberg, Datastream, Kenneth French, Goldman Sachs Global Investment Research

Tech boom has led to a higher concentration in a few large cap stocks

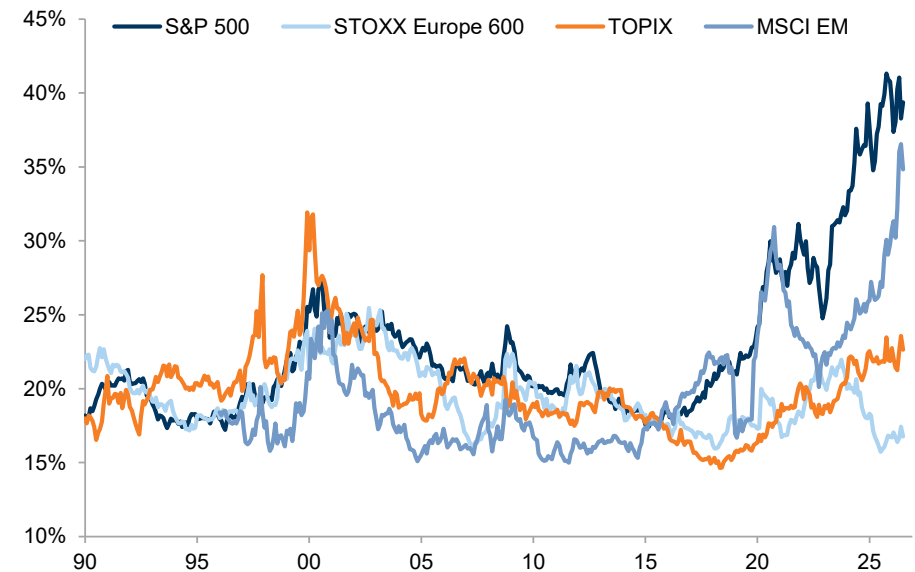
Concentration in the US equity indices has increased alongside the Tech sector weight

Weight in US equity benchmark index



Stock concentration has increased both in the US and, more recently, in EM benchmarks

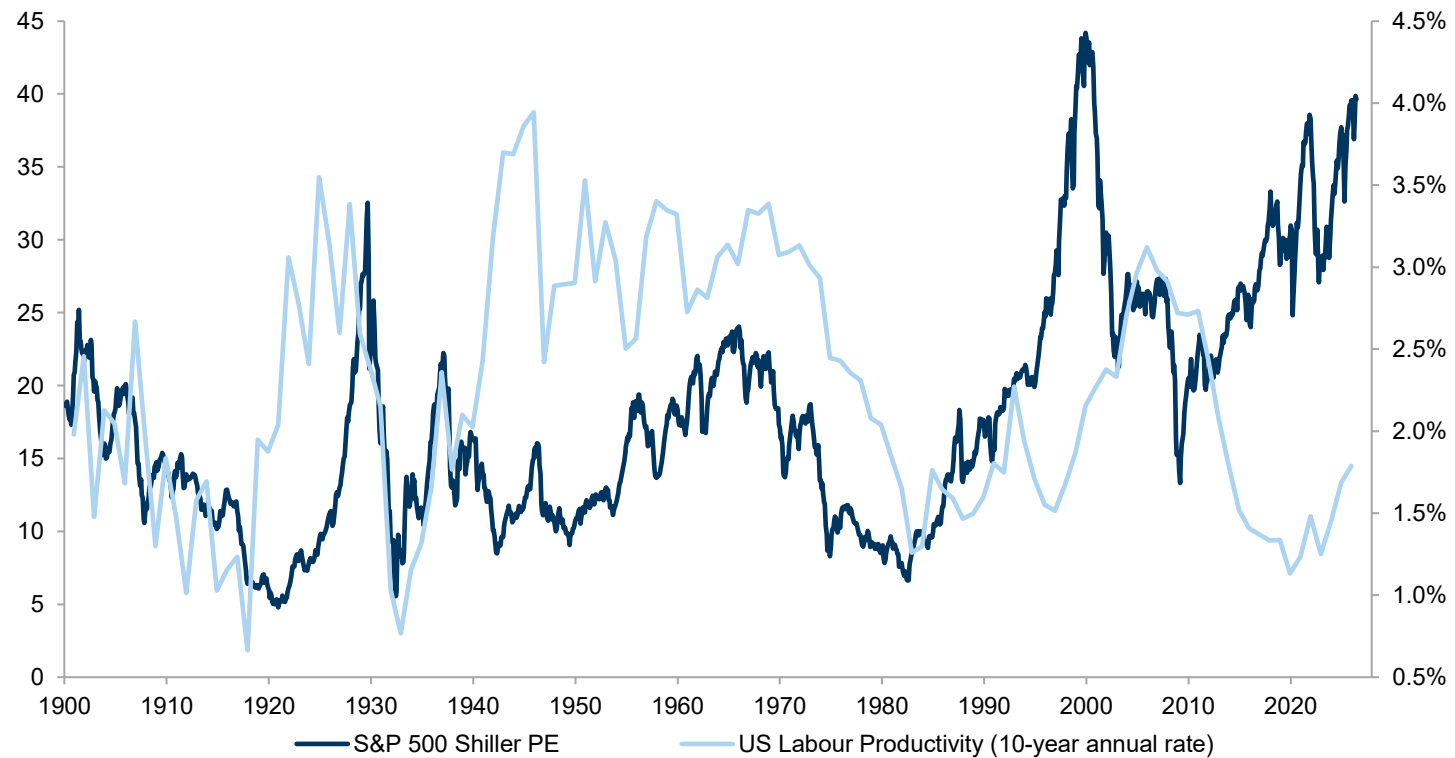
Weight in the largest 10 stocks



Source: Datastream, Goldman Sachs Global Investment Research

S&P 500 Shiller P/Es are nearing Tech Bubble levels without much productivity growth so far

S&P 500 Shiller P/E vs. US labour productivity growth



Source: Robert Shiller, Haver Analytics, Goldman Sachs Global Investment Research

US equity valuations are elevated but supported by macro conditions and high corporate profitability...so far

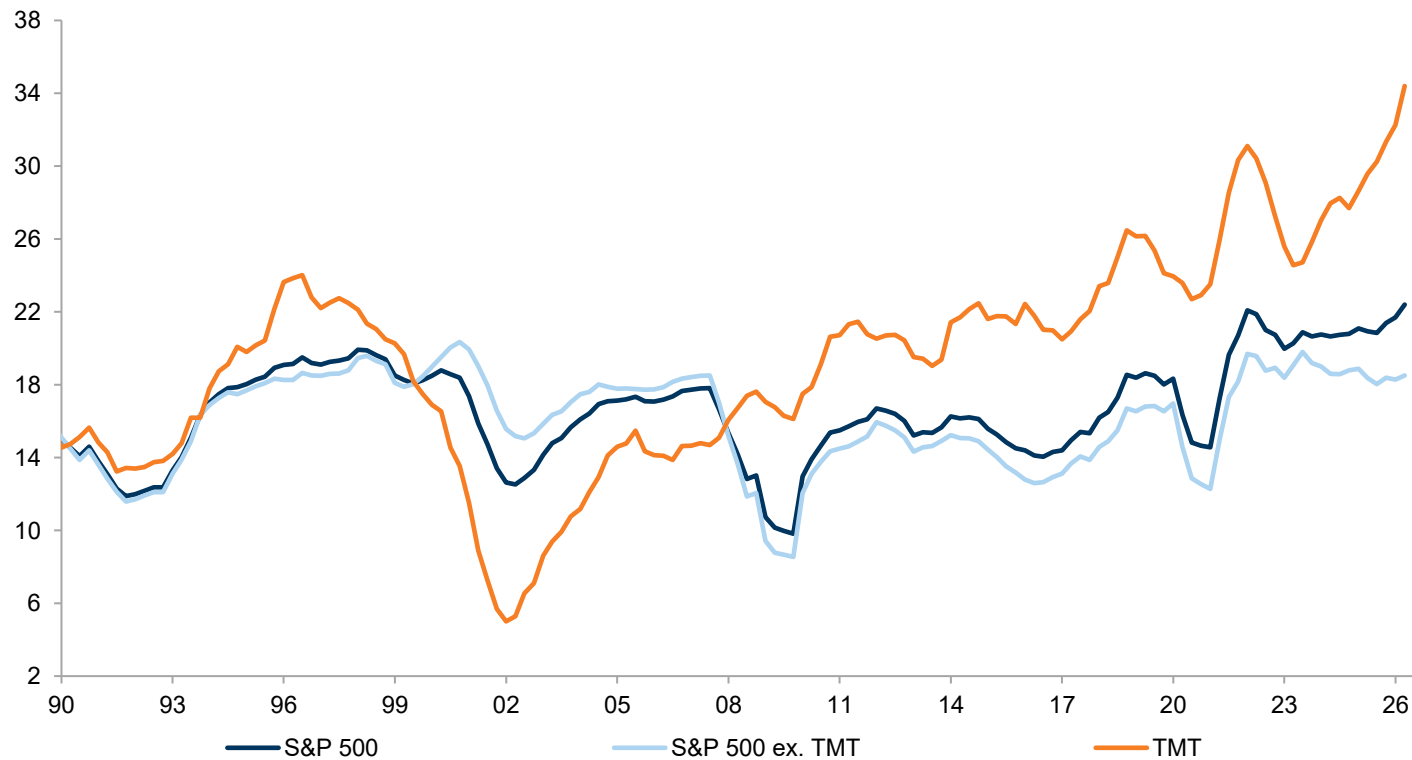
US equity valuations are not much above their structural fair value assuming normalised inflation and current ROE



Source: Haver Analytics, Kenneth French, Robert Shiller, Goldman Sachs Global Investment Research

S&P 500 profitability has been boosted by the TMT sector

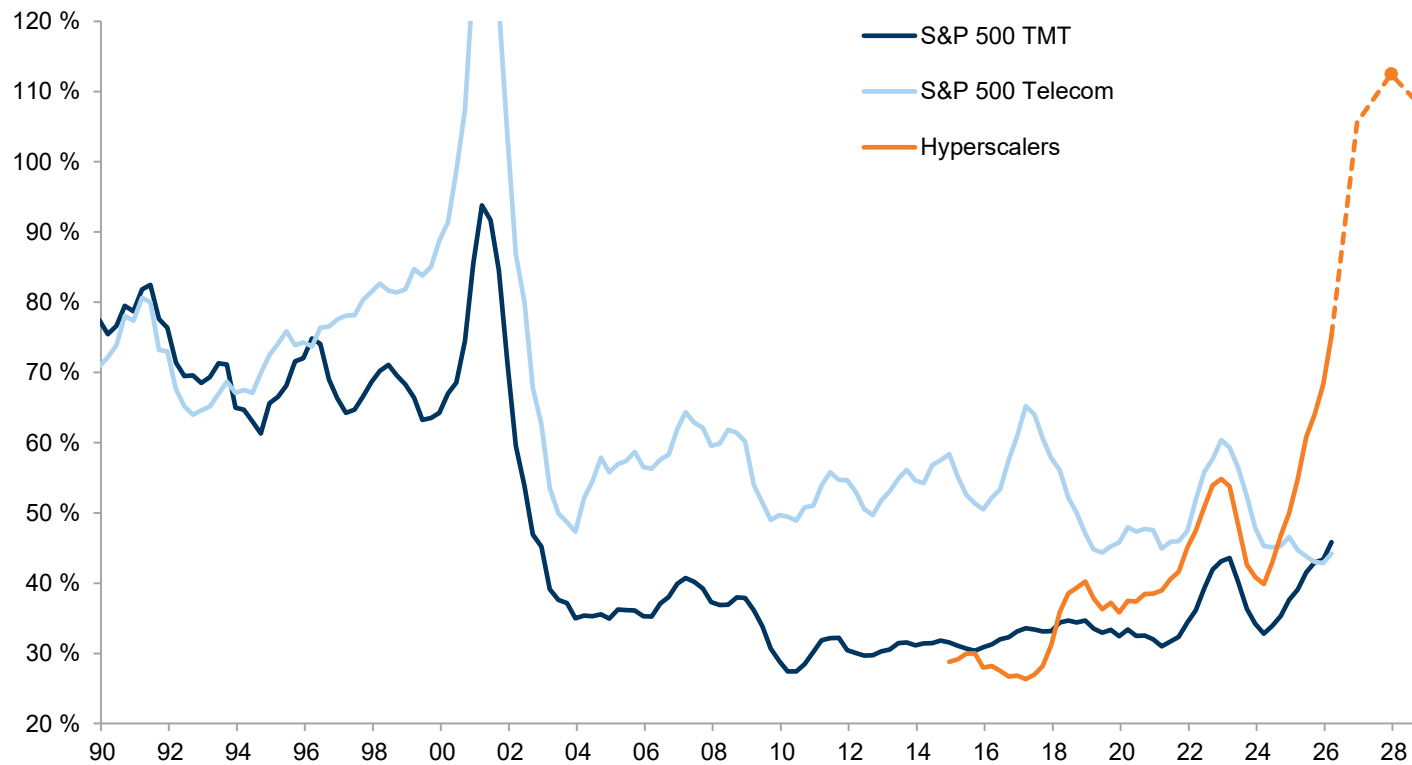
Return on equity (ROE, %)



Source: Compustat, Goldman Sachs Global Investment Research

AI capex has been weighing on FCF from the hyperscalers

Capex as a % of cash flow from operations, dashed line represents 2026 consensus

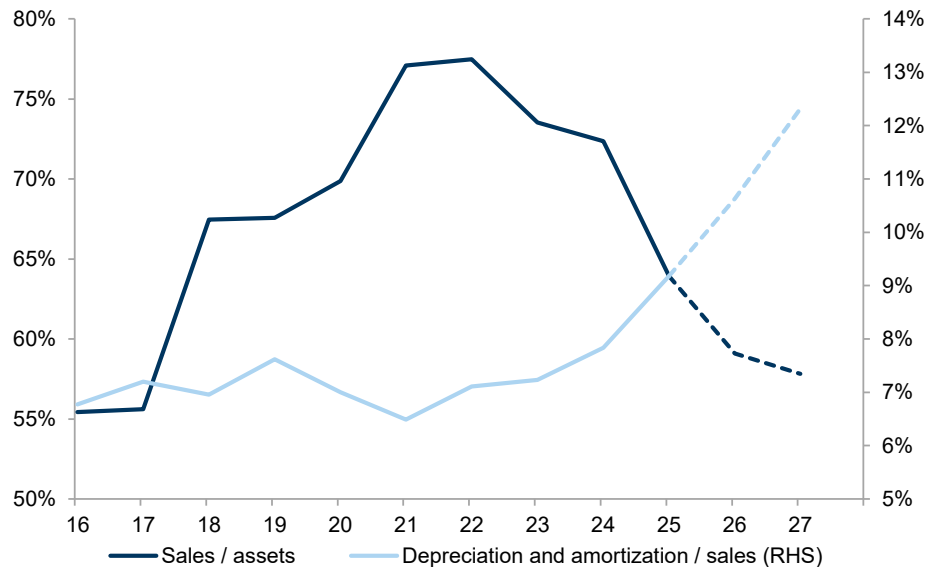


Source: Compustat, Factset, Goldman Sachs Global Investment Research

The outlook for profitability among the hyperscalers is becoming more challenging

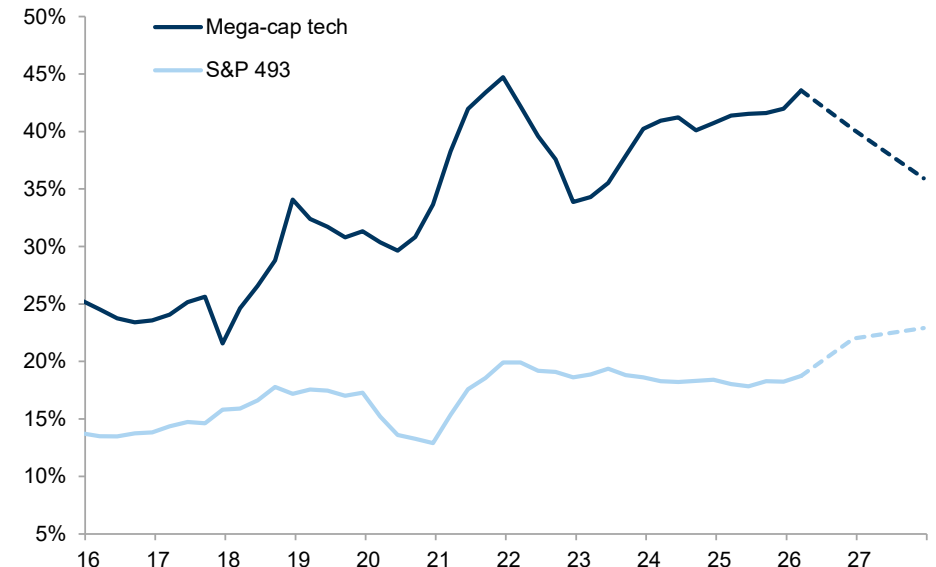
AI hyperscaler capex creates headwinds to ROE

Hyperscalers = AMZN, META, MSFT, GOOGL, ORCL



ROEs for Mega-cap tech stocks are expected to decline

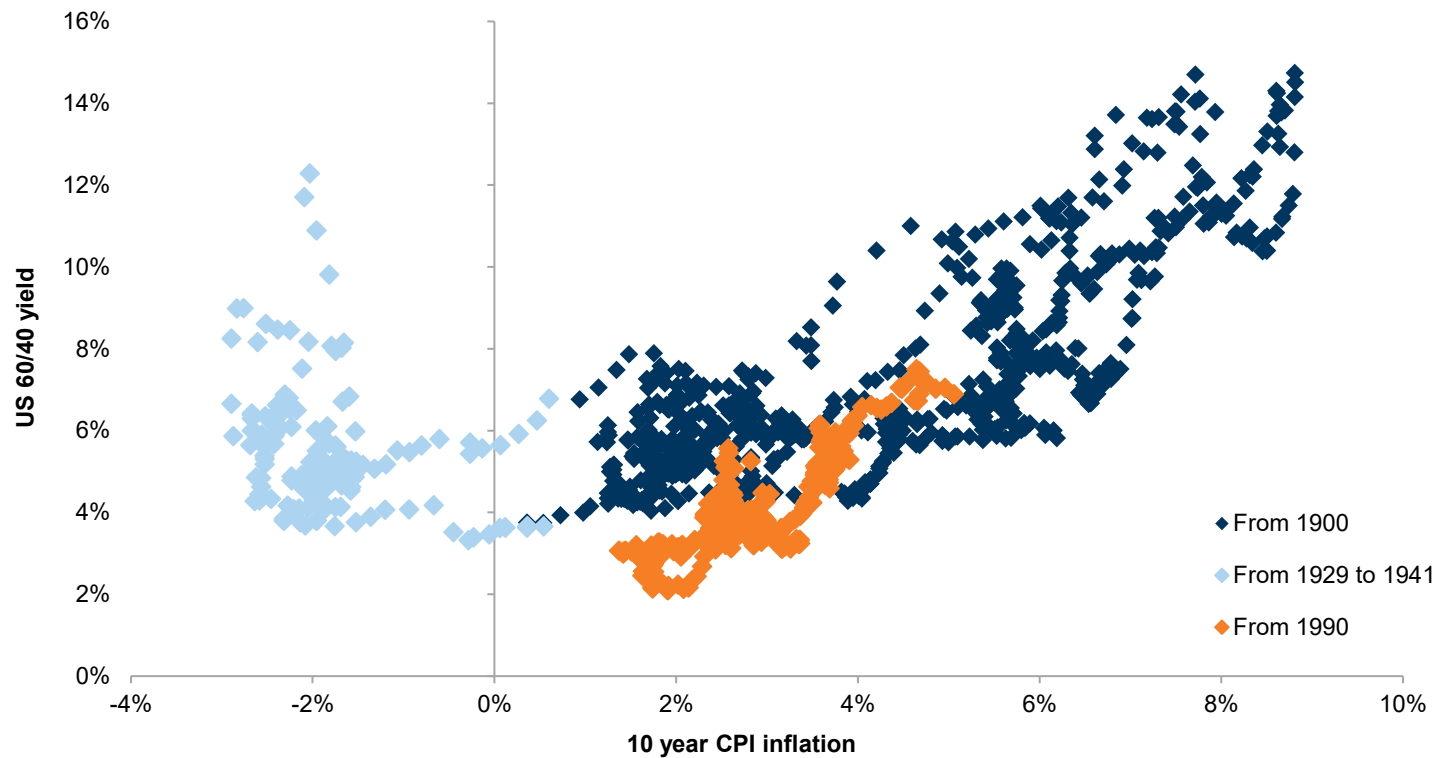
Mega-cap tech = AAPL, AMZN, AVGO, GOOGL, META, MSFT, NVDA



Source: Compustat, FactSet, Goldman Sachs Global Investment Research

Anchored inflation near 2% most supportive for valuations – higher inflation volatility would be negative

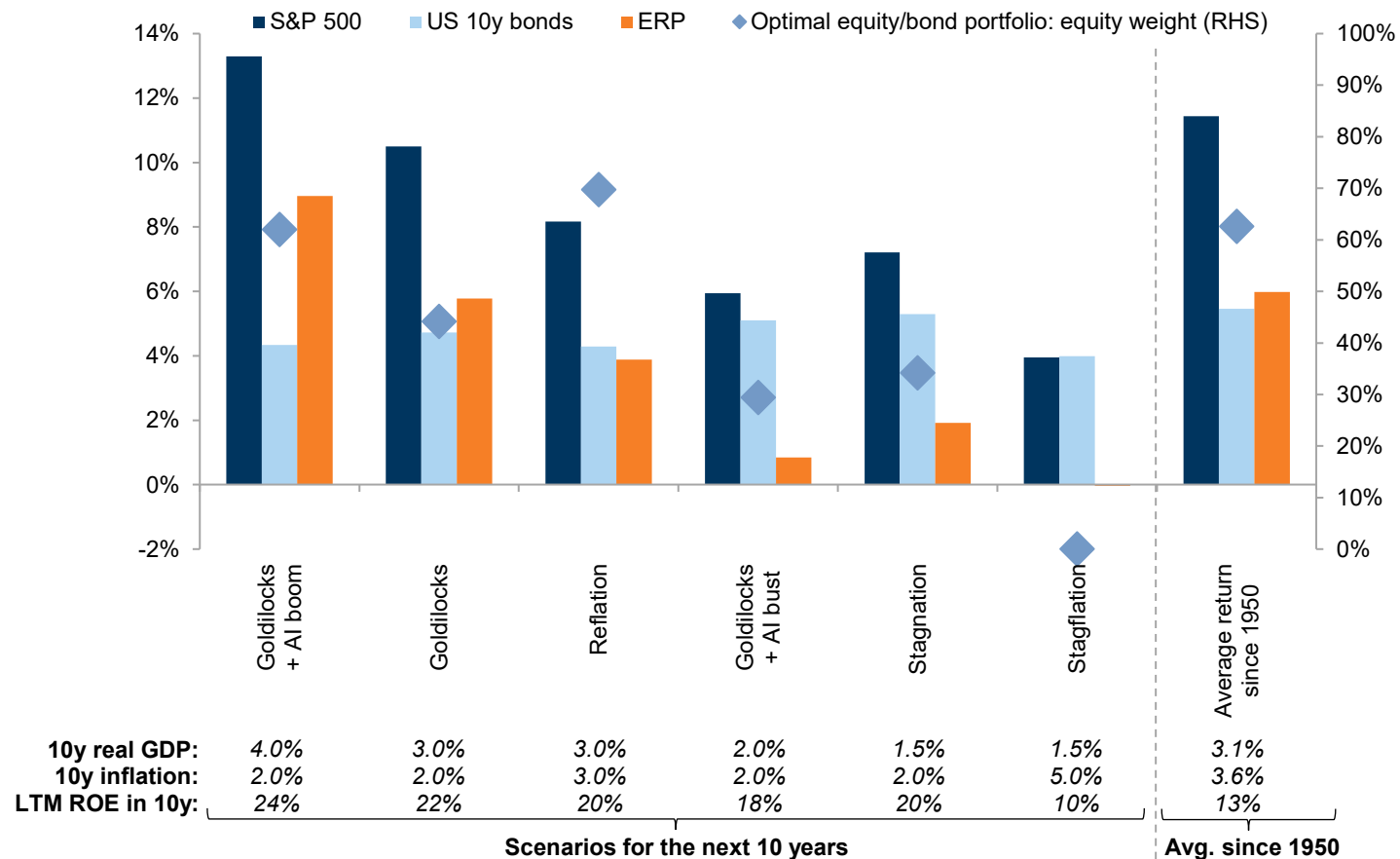
US 60/40 yield = 60% S&P 500 Shiller earnings yield + 40% US 10-year yield



Source: Robert Shiller, Goldman Sachs Global Investment Research

Structural regimes point to very different long-term return forecasts

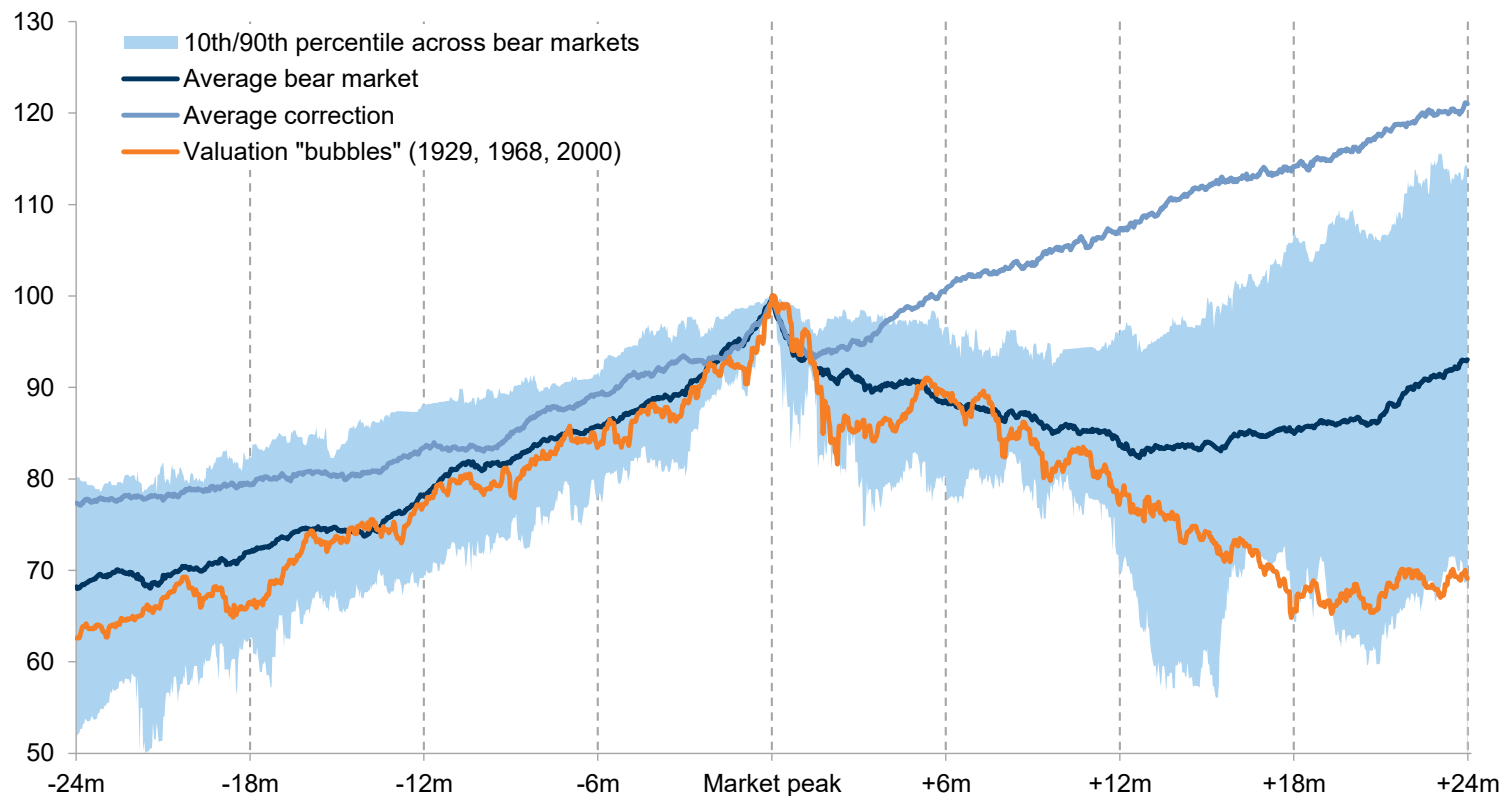
10-year return scenarios in function of different structural scenarios



Source: Haver Analytics, Robert Shiller, Goldman Sachs Global Investment Research

Being underinvested late cycle can be costly, with equities usually performing well up until the bull market peak

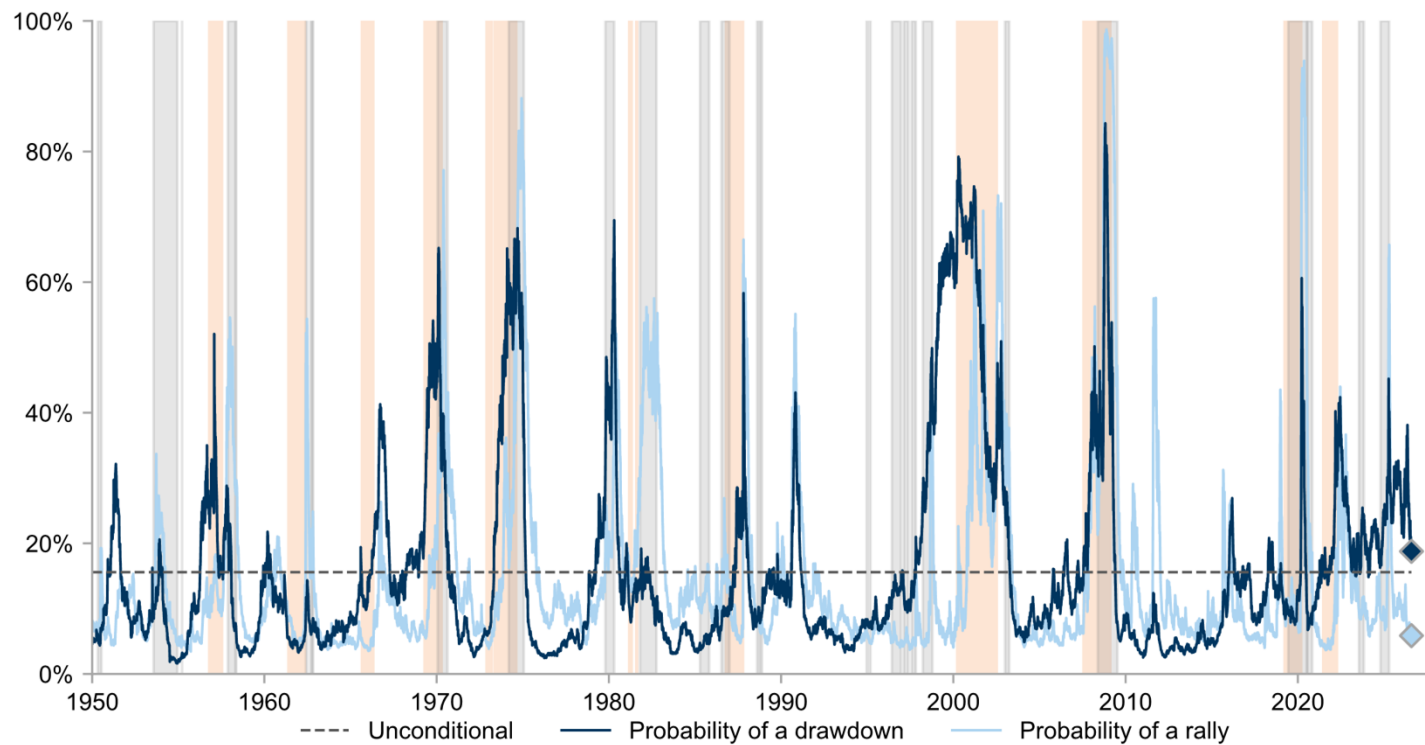
Average S&P 500 total return performance around bear markets and corrections (data since 1900, 21 bear markets, 30 corrections included)



Source: Datastream, Haver Analytics, Goldman Sachs Global Investment Research

Our equity tail risk framework points to a less negative asymmetry....

Probability of S&P 500 drawdown/rally based on multi-variate logit model (Orange/ light grey shading = S&P 500 subsequent drawdown >-20%/ rally >+35%)



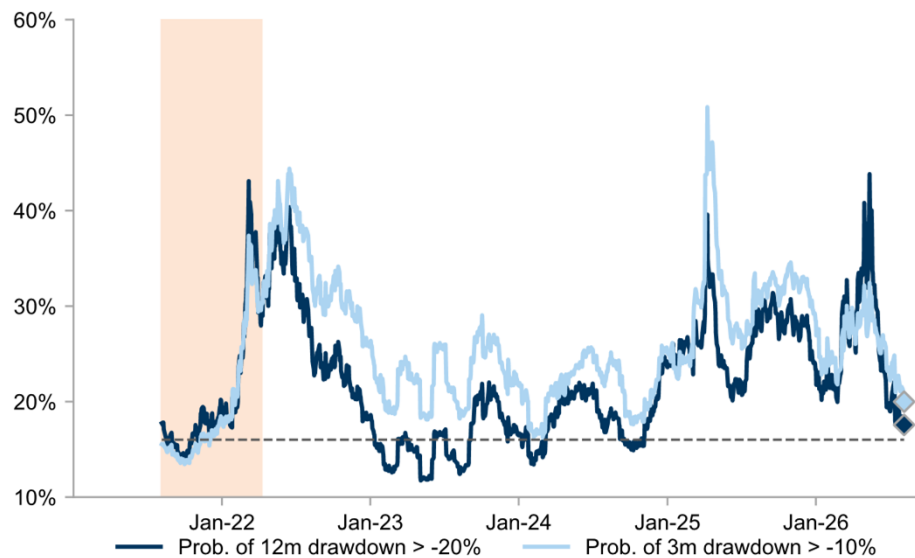
Source: Haver Analytics, Datastream, Goldman Sachs Global Investment Research

...but with conflicting signals from valuations and the business cycle

Implied probability of S&P drawdown based on multi-variate logit model

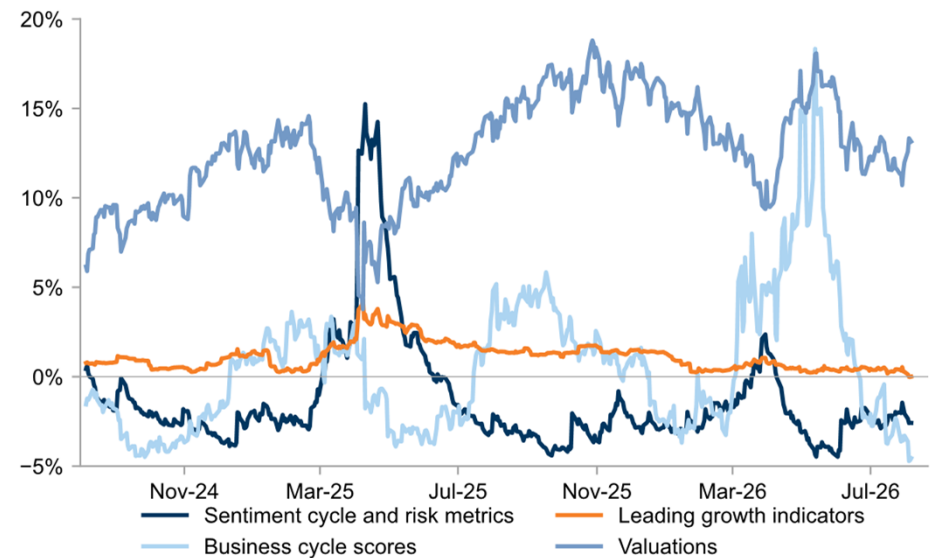
Orange shading = S&P 500 subsequent drawdown >20%.

Dashed line = unconditional probability



Contribution to the probability of a drawdown >20% in 12m

Based on Shapley values. Inputs aggregated by type



Source: Haver Analytics, Datastream, Goldman Sachs Global Investment Research

Real assets have done well during Tech-led equity drawdowns

Performance during TMT drawdowns of more than 10% and 5% underperformance vs. the S&P 500

Peak	Trough	Length (months)	TMT drawdown	S&P 500	US 10y	Gold	S&P GSCI	Real Asset Stocks	Real Asset Index
May-50	Jul-50	2	-13%	-5%	0%		1%	-4%	
Aug-60	Oct-60	2	-10%	-6%	0%	3%	1%	-4%	-2%
Nov-61	Jun-62	7	-38%	-23%	2%	0%	-2%	-21%	-13%
Dec-67	Feb-68	2	-13%	-7%	2%	1%	3%	-4%	-2%
May-68	Jun-70	25	-39%	-26%	0%	-17%	18%	-18%	-10%
Jan-73	Sep-74	20	-52%	-45%	2%	129%	132%	-43%	5%
Aug-78	Oct-78	2	-14%	-10%	-2%	16%	7%	-8%	-1%
Jan-80	Mar-80	2	-16%	-11%	-7%	-26%	-15%	-11%	-15%
Jun-83	Jul-84	13	-22%	-10%	2%	-19%	5%	-4%	-5%
May-86	Sep-86	4	-12%	-6%	7%	24%	11%	4%	9%
Aug-87	Aug-90	36	-37%	-2%	31%	-15%	114%	13%	24%
Mar-00	Sep-02	30	-79%	-46%	34%	16%	16%	1%	8%
Dec-21	Sep-22	9	-33%	-25%	-17%	-8%	22%	-6%	-1%
Dec-24	Mar-25	3	-11%	-5%	4%	19%	5%	7%	9%
Oct-25	Mar-26	5	-11%	-5%	0%	16%	39%	18%	22%
Average		10.8	-26.7%	-15.5%	3.9%	9.8%	23.7%	-5.4%	2.0%
Median		5.0	-15.8%	-9.8%	1.8%	2.1%	6.7%	-4.4%	-0.8%

Source: Datastream, Haver Analytics, Goldman Sachs Global Investment Research

Low vol, high dividend yield, value stocks outperform in TMT drawdowns

Performance during TMT drawdowns of more than 10% and 5% underperformance vs. the S&P 500

Peak	Trough	Length (months)	TMT drawdown	S&P 500	US 10y	S&P low vol	High DY	Value vs Growth	Cycl vs Defs
May-50	Jul-50	2	-13%	-5%	0%		-4%	16%	5%
Aug-60	Oct-60	2	-10%	-6%	0%		-5%	2%	-3%
Nov-61	Jun-62	7	-38%	-23%	2%		-9%	12%	-4%
Dec-67	Feb-68	2	-13%	-7%	2%		-1%	6%	-3%
May-68	Jun-70	25	-39%	-26%	0%		-12%	7%	-4%
Jan-73	Sep-74	20	-52%	-45%	2%		-31%	43%	10%
Aug-78	Oct-78	2	-14%	-10%	-2%		-6%	0%	0%
Jan-80	Mar-80	2	-16%	-11%	-7%	-10%	-8%	-1%	-3%
Jun-83	Jul-84	13	-22%	-10%	2%	0%	6%	25%	-9%
May-86	Sep-86	4	-12%	-6%	7%	-1%	5%	14%	-2%
Aug-87	Aug-90	36	-37%	-2%	31%	1%	15%	-2%	-23%
Mar-00	Sep-02	30	-79%	-46%	34%	16%	30%	79%	-19%
Dec-21	Sep-22	9	-33%	-25%	-17%	-16%	-4%	22%	-5%
Dec-24	Mar-25	3	-11%	-5%	4%	7%	7%	16%	-5%
Oct-25	Mar-26	5	-11%	-5%	0%	4%	17%	17%	5%
Average		10.8	-26.7%	-15.5%	3.9%	0.1%	-0.1%	17.0%	-4.0%
Median		5.0	-15.8%	-9.8%	1.8%	0.8%	-4.1%	14.1%	-3.2%

Source: Datastream, Haver Analytics, Goldman Sachs Global Investment Research

Non-US equities tend to outperform during Tech-led equity sell-offs but FX performance is mixed

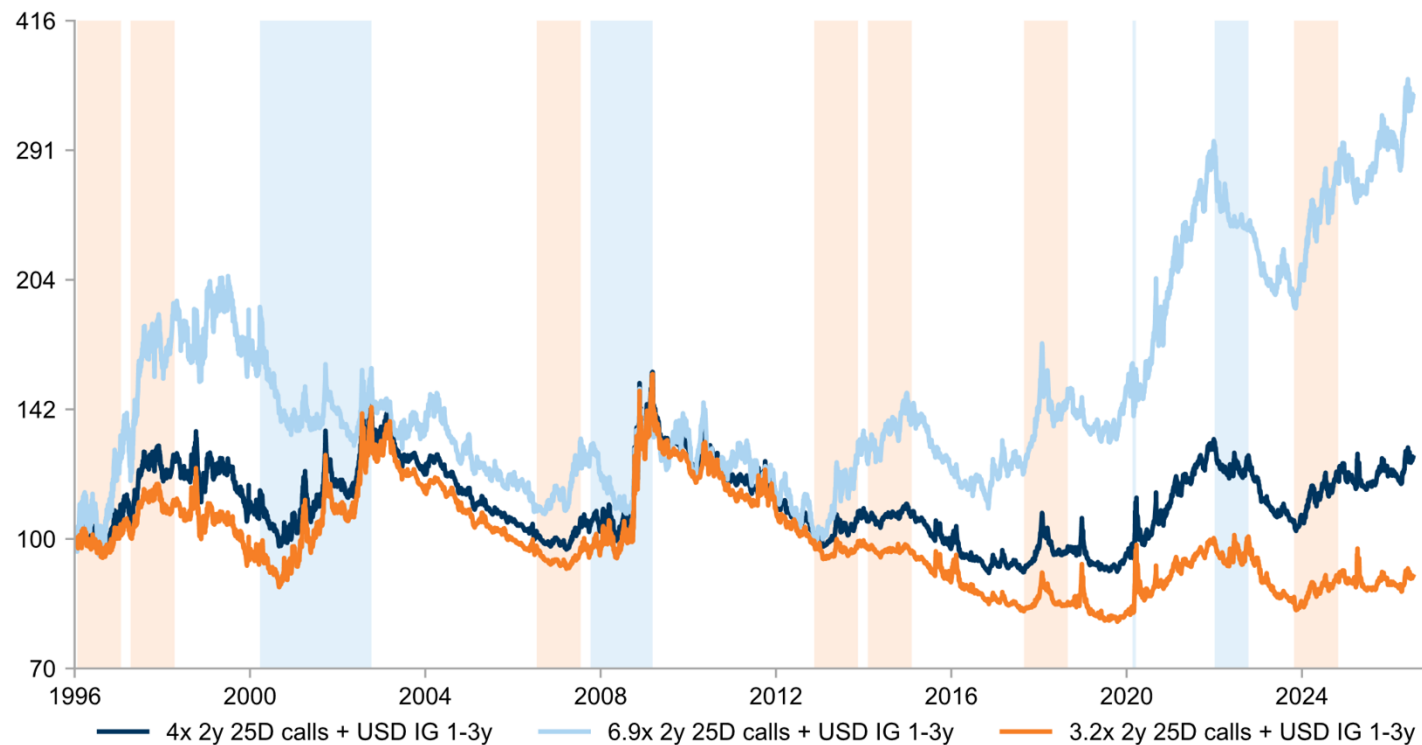
Performance during TMT drawdowns of more than 10% and 5% underperformance vs. the S&P 500

Peak	Trough	Length (months)	TMT drawdown	S&P 500	US 10y	CDAX	TOPIX	MSCI EM	CHF	EUR	JPY
May-50	Jul-50	2	-13%	-5%	0%	2%	4%	-2%			
Aug-60	Oct-60	2	-10%	-6%	0%	-6%	-	2%			
Nov-61	Jun-62	7	-38%	-23%	2%	-25%	6%	-2%			
Dec-67	Feb-68	2	-13%	-7%	2%	6%	4%	9%			
May-68	Jun-70	25	-39%	-26%	0%	-9%	42%	18%	0%	2%	0%
Jan-73	Sep-74	20	-52%	-45%	2%	-30%	-30%	-16%	22%	6%	2%
Aug-78	Oct-78	2	-14%	-10%	-2%	0%	3%	0%	9%	5%	6%
Jan-80	Mar-80	2	-16%	-11%	-7%	-5%	-3%	1%	-12%	-9%	-4%
Jun-83	Jul-84	13	-22%	-10%	2%	2%	16%	5%	-15%	-13%	-2%
May-86	Sep-86	4	-12%	-6%	7%	1%	17%	-5%	17%	10%	13%
Aug-87	Aug-90	36	-37%	-2%	31%	7%	-8%	57%	15%	15%	-1%
Mar-00	Sep-02	30	-79%	-46%	34%	-64%	-46%	-47%	13%	3%	-16%
Dec-21	Sep-22	9	-33%	-25%	-17%	-30%	-8%	-29%	-7%	-14%	-20%
Dec-24	Mar-25	3	-11%	-5%	4%	11%	-5%	2%	2%	4%	5%
Oct-25	Mar-26	5	-11%	-5%	0%	-4%	5%	0%	0%	0%	-3%
Average		10.8	-26.7%	-15.5%	3.9%	-9.6%	-0.2%	-0.4%	4.0%	0.8%	-1.9%
Median		5.0	-15.8%	-9.8%	1.8%	-4.0%	3.4%	-0.3%	2.4%	3.2%	-0.9%

Source: Datastream, Haver Analytics, Goldman Sachs Global Investment Research

Long-dated equity calls can improve asymmetry or increase upside convexity

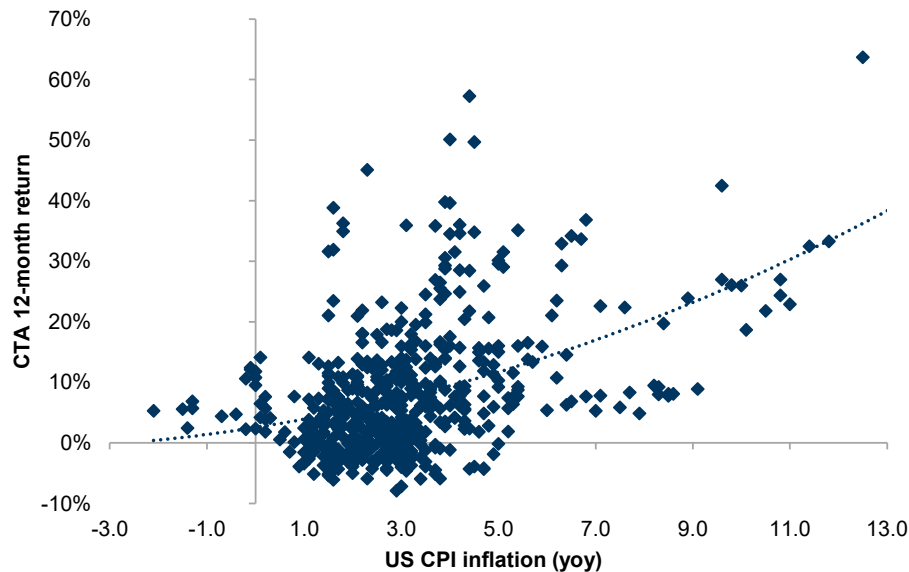
Indexed total return of a portfolio of 2-year calls on S&P 500 (rolled after 1 year) + USD 1-3y IG vs. S&P 500. Orange shade: late-cycle rallies. Blue shade: Bear markets



Source: Goldman Sachs FICC and Equities, Goldman Sachs Global Investment Research

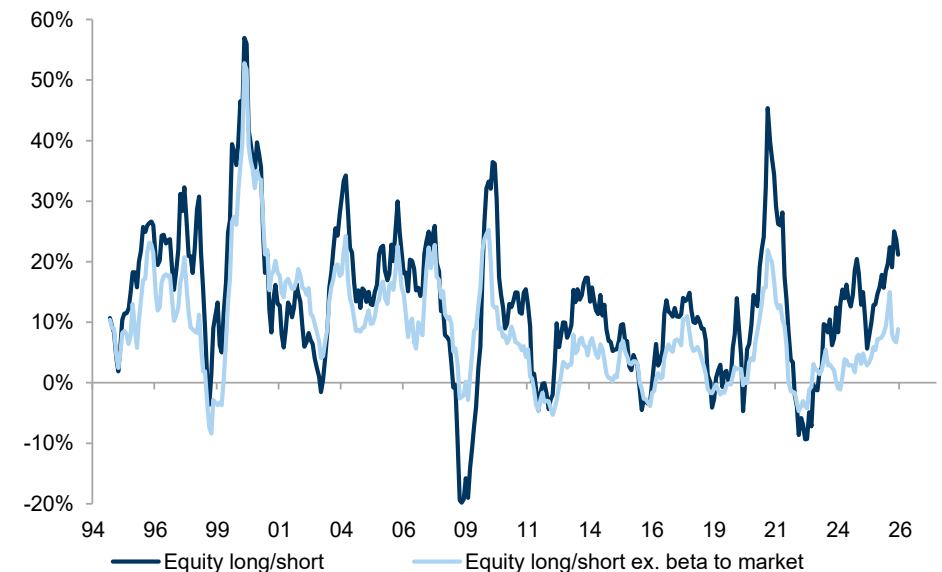
During periods of elevated inflation, CTAs have often delivered stronger returns – equity/long HFs short can benefit from dispersion

Higher levels of inflation tend to be more supportive of CTA performance



Equity long/short hedge funds did well during the Tech Bubble, also in excess of the market beta

12-month rolling performance and excluding beta to the market

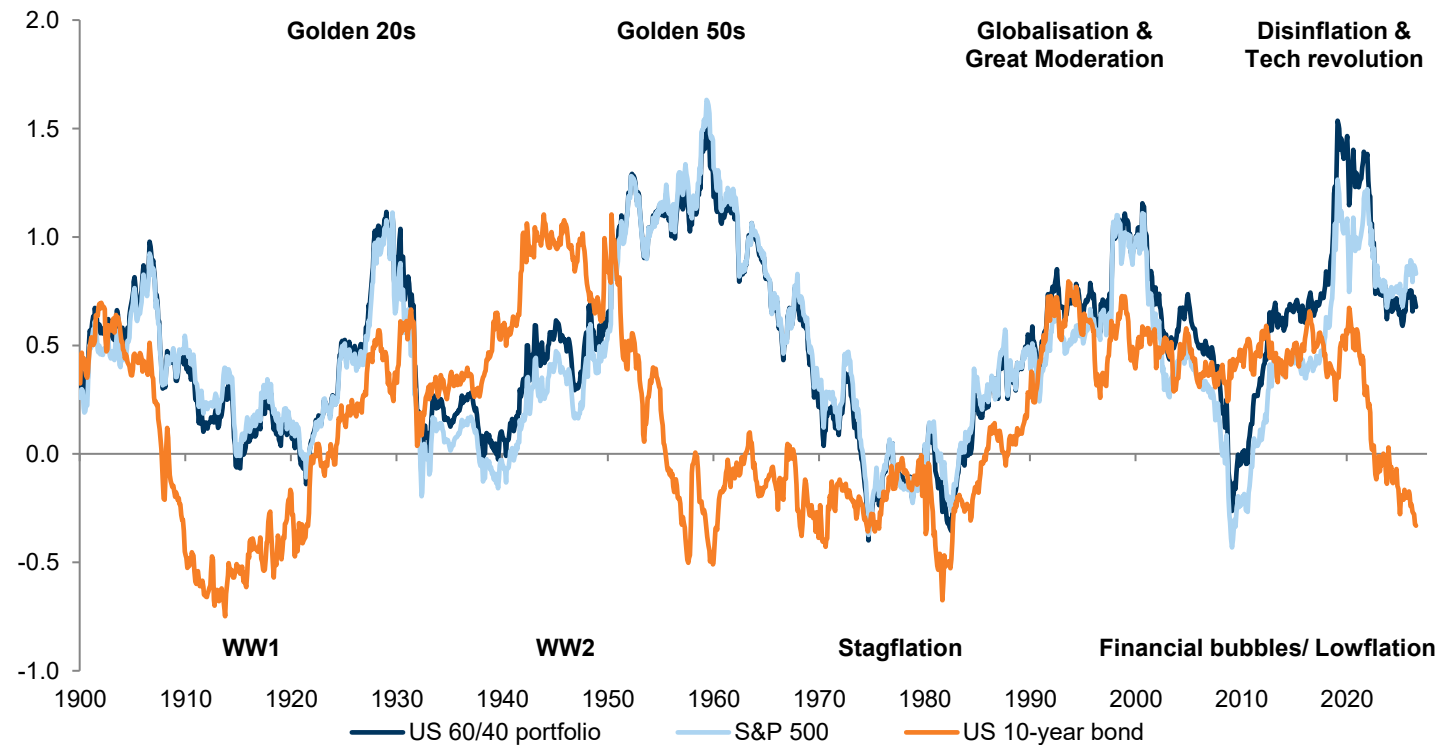


Source: Preqin, Datastream, Haver Analytics, Bloomberg, Goldman Sachs Global Investment Research

Long-Term Portfolio Construction

US 60/40 portfolio performance varied materially owing to structural cycles linked to innovation and inflation

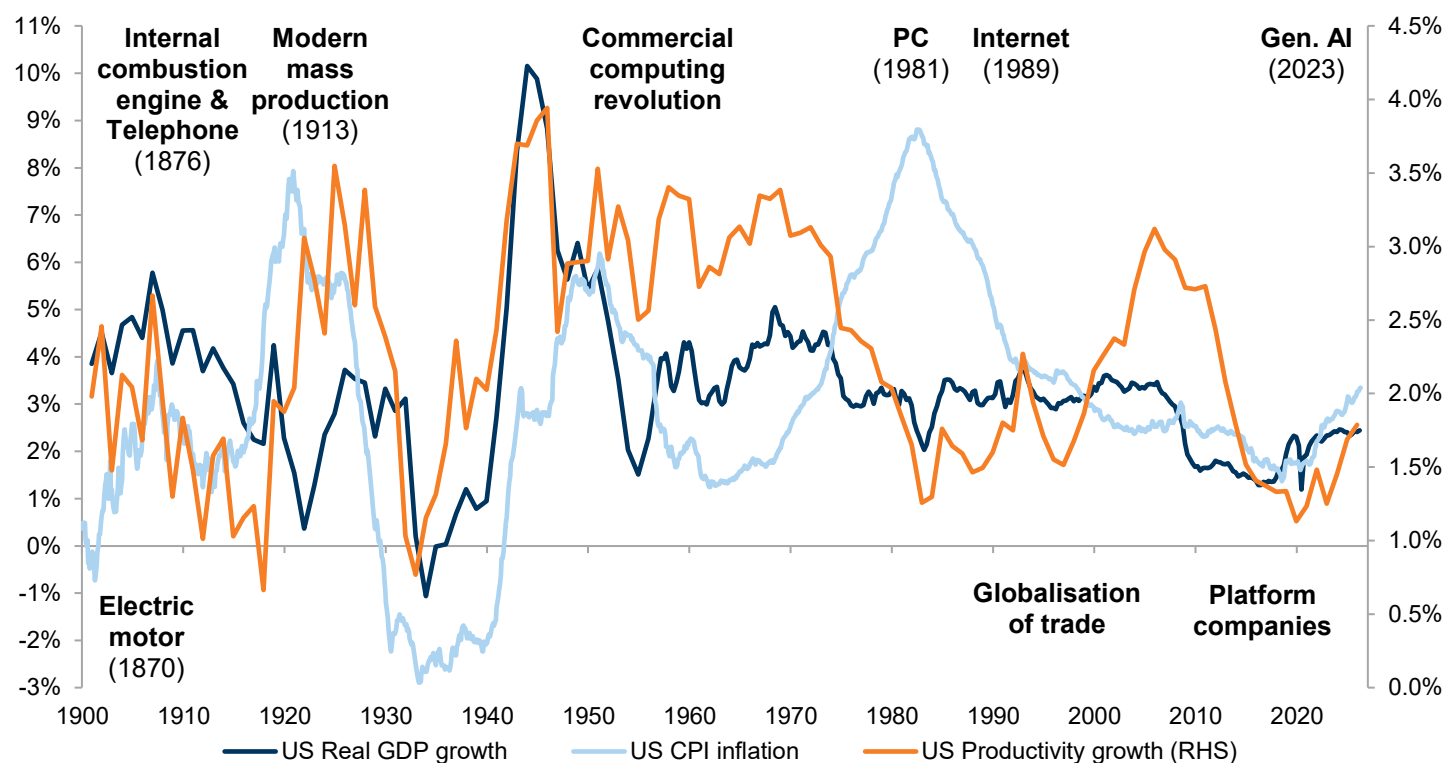
10-year rolling Sharpe ratio



Source: Haver Analytics, Robert Shiller, Goldman Sachs Global Investment Research

Innovation was historically a key driver of structural cycles

10-year rolling average of US GDP growth, CPI inflation and labour productivity growth

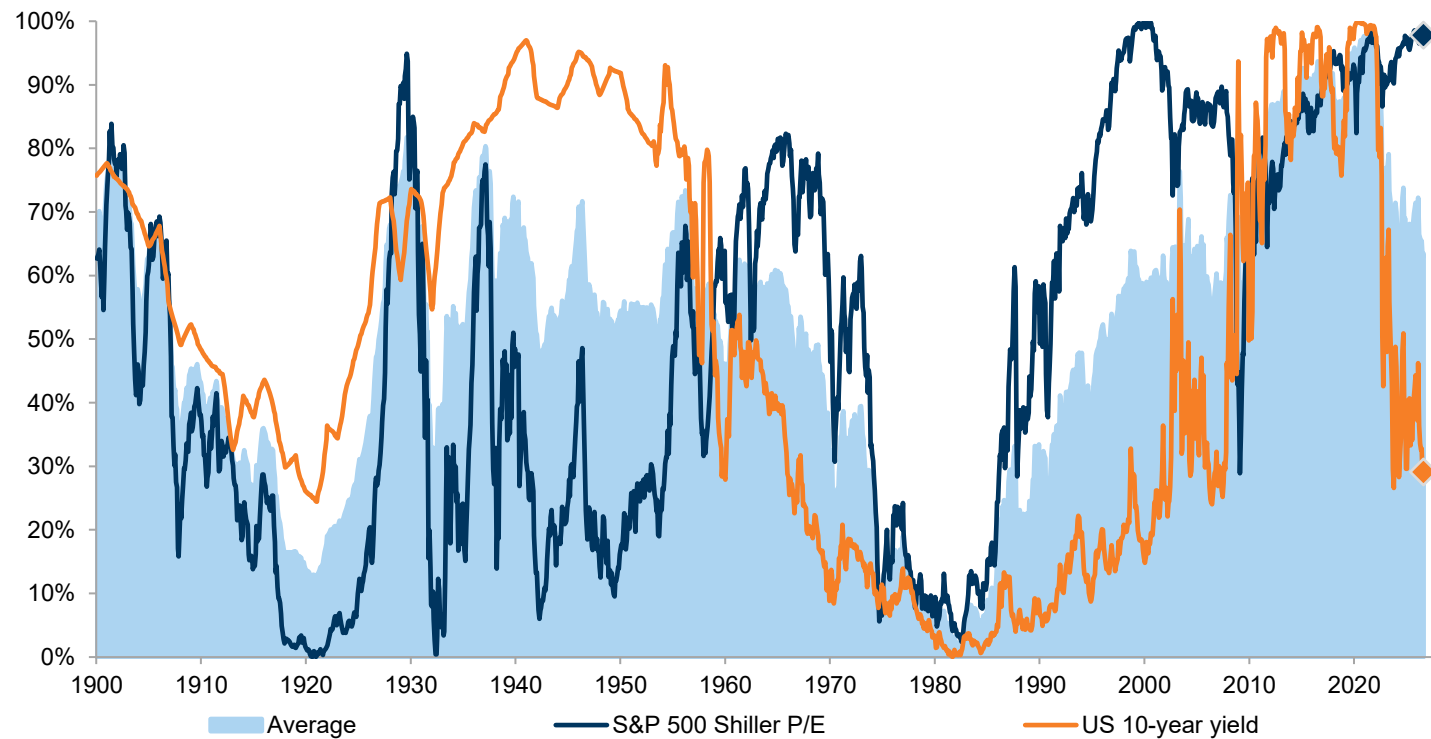


Source: Robert Shiller, Haver Analytics, Goldman Sachs Global Investment Research

Less valuation frustration in bonds but equities are again expensive

While equities remain relatively expensive, bond yields are now close to long-run average

Valuation percentile (since 1871). A higher percentile means valuations are more expensive

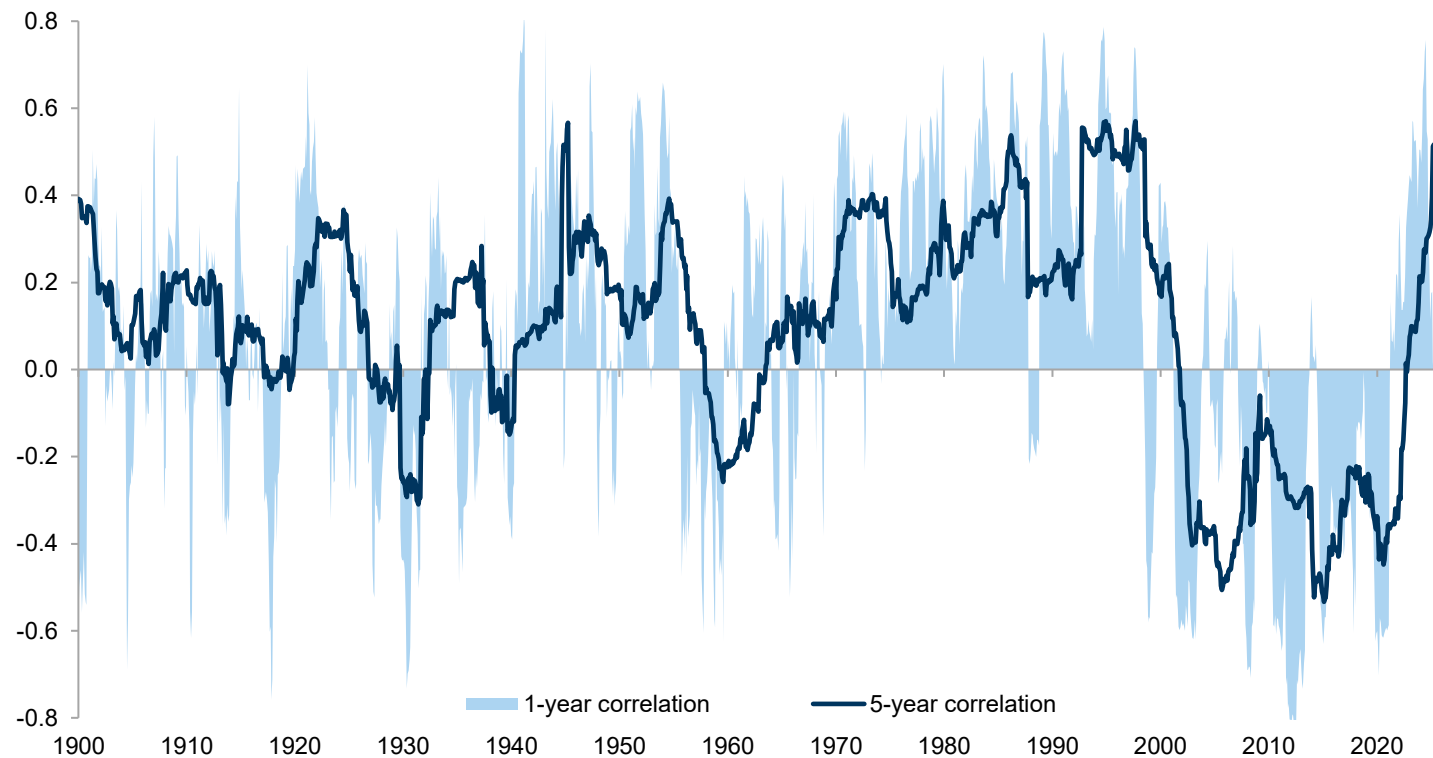


Source: Robert Shiller, Haver Analytics, Goldman Sachs Global Investment Research

Since the COVID-19 crisis equity/bond correlations have been more positive

S&P 500 vs. US 10-Year bond correlation

Daily returns where available, otherwise monthly returns



Source: Haver Analytics, Robert Shiller, Goldman Sachs Global Investment Research

Market-value weighted benchmarks are not necessarily a good starting point for both active and passive multi-asset portfolios

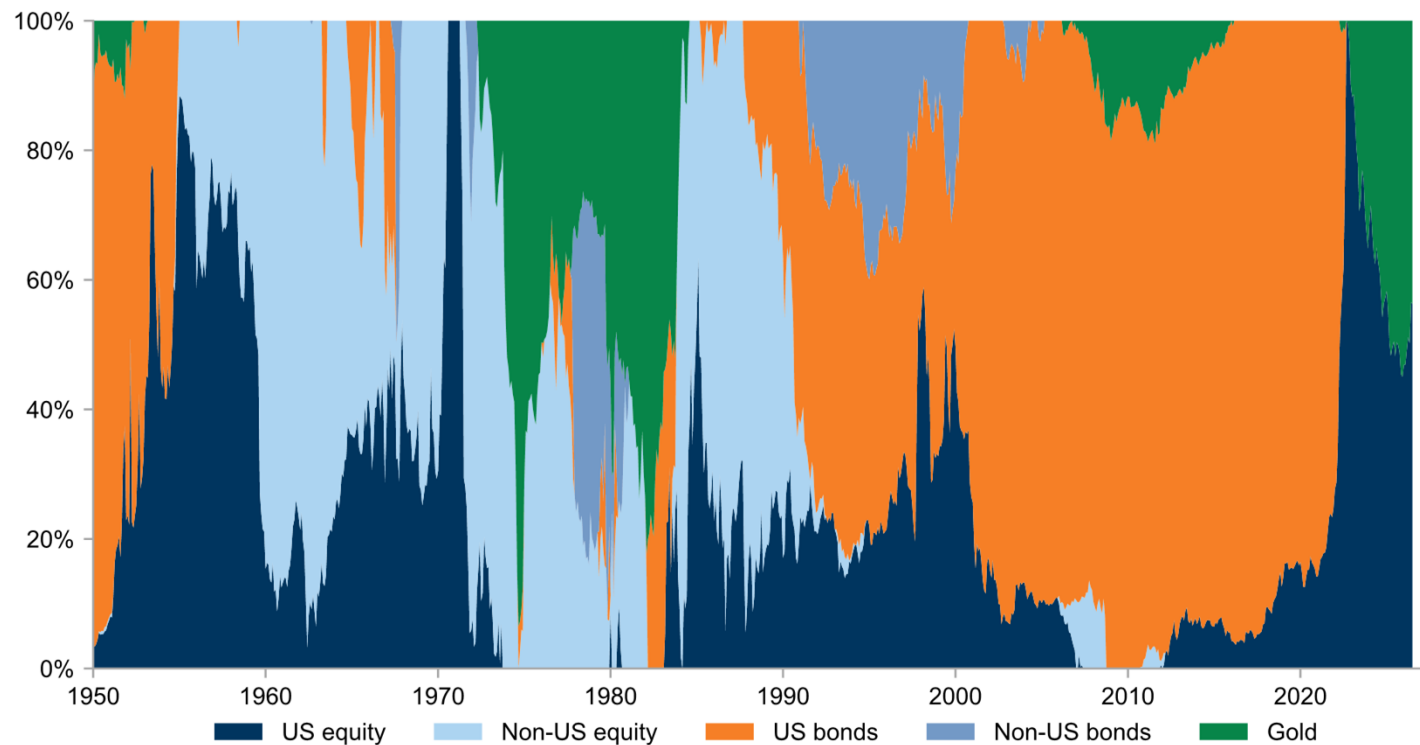
Performance statistics (based on monthly returns, data since 1950)

	World portfolio	60/40 portfolio	Global risk parity	Global equities	Global bonds	Gold
Since 1950						
Return (p.a.)	7.8%	9.0%	7.2%	11.1%	5.3%	6.2%
Volatility	8.3%	9.2%	6.9%	13.6%	5.6%	17.0%
Sharpe ratio	0.43	0.51	0.44	0.50	0.21	0.12
5% CVaR (monthly)	-5.0%	-5.6%	-3.8%	-8.8%	-3.0%	-9.2%
95% CVaR (monthly)	5.7%	6.5%	5.2%	8.9%	4.4%	13.7%
Real return (p.a.)	4.1%	5.3%	3.6%	7.3%	1.8%	2.5%
Lowest ann. return	-27%	-33%	-18%	-48%	-21%	-38%
Highest ann. return	50%	57%	41%	72%	44%	179%
Since 1990						
Return (p.a.)	6.4%	7.1%	6.2%	8.2%	4.7%	6.4%
Volatility	9.3%	10.3%	7.7%	15.0%	6.6%	17.3%
Sharpe ratio	0.39	0.42	0.45	0.36	0.30	0.21
5% CVaR (monthly)	-5.7%	-6.5%	-4.1%	-10.1%	-3.4%	-8.4%
95% CVaR (monthly)	5.5%	6.5%	4.7%	9.2%	3.8%	10.7%
Real return (p.a.)	3.7%	4.3%	3.5%	5.5%	2.0%	3.7%
Lowest ann. return	-27%	-33%	-18%	-48%	-21%	-27%
Highest ann. return	30%	39%	28%	59%	21%	58%

Source: Bloomberg, Datastream, Goldman Sachs Global Investment Research

Over 10-year rolling horizons, the optimal portfolio shifted notably over time, deviating materially from the World Portfolio benchmark

Average of rolling weight of highest Sharpe ratio portfolio (data since 1950)



Source: Bloomberg, Datastream, Goldman Sachs Global Investment Research

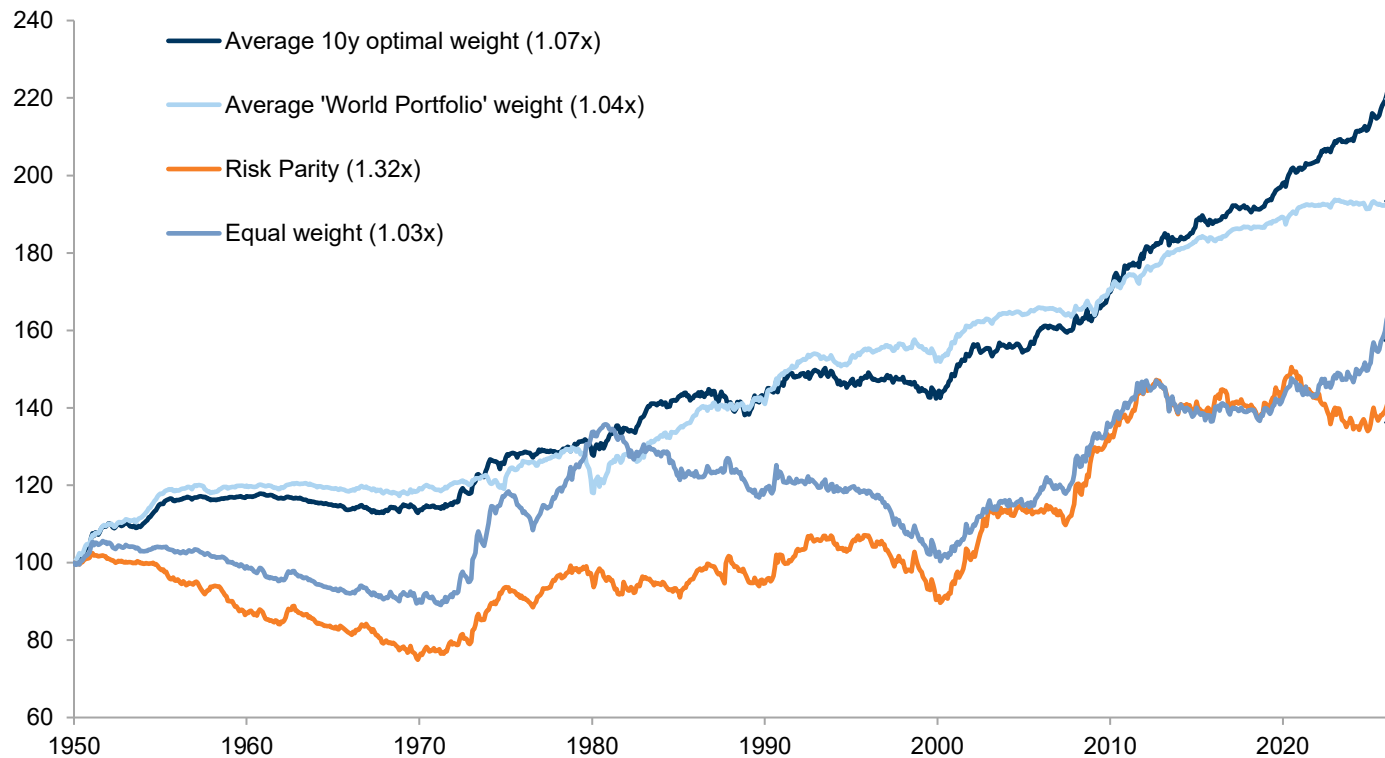
A 'neutral' global multi-asset benchmark would be very different to the current World Portfolio (for a US\$ investor)

	World Portfolio	Average 10y optimal weight	Average 'World Portfolio' weight	Risk Parity	Equal weight
Asset weights					
	<i>Current</i>				
US equity	35%	24%	25%	13%	20%
Non-US equity	20%	23%	24%	13%	20%
US bonds	19%	38%	30%	40%	20%
Non-US bonds	19%	5%	18%	23%	20%
Gold	6%	11%	4%	11%	20%
Performance metrics since 1950					
Annualised return	7.9%	8.7%	8.7%	7.4%	8.4%
Annualised volatility	8.2%	7.6%	7.8%	6.2%	7.9%
Tracking error		1.7%	1.4%	3.3%	2.8%
Sharpe ratio	0.45	0.58	0.56	0.51	0.53
Information ratio		0.44	0.50	-0.16	0.17
5% CVaR of 1m returns	-5.0%	-4.5%	-4.7%	-3.4%	-4.5%
1y relative return	5th perc.	-1.9%	-1.4%	-6.2%	-4.2%
	95th perc.	4.1%	4.0%	7.4%	6.6%
10y relative return	5th perc.	-0.4%	-0.1%	-2.6%	-1.5%
	95th perc.	1.6%	1.5%	1.8%	3.2%
Hit ratio for real returns since 1950					
1y rolling real return	above 5%	50%	54%	53%	48%
	below 0%	24%	23%	23%	25%
10y rolling real return	above 5%	35%	40%	26%	37%
	below 0%	5%	3%	1%	0%

Source: Bloomberg, Datastream, Goldman Sachs Global Investment Research

Regime-neutral portfolios outperformed in the medium term

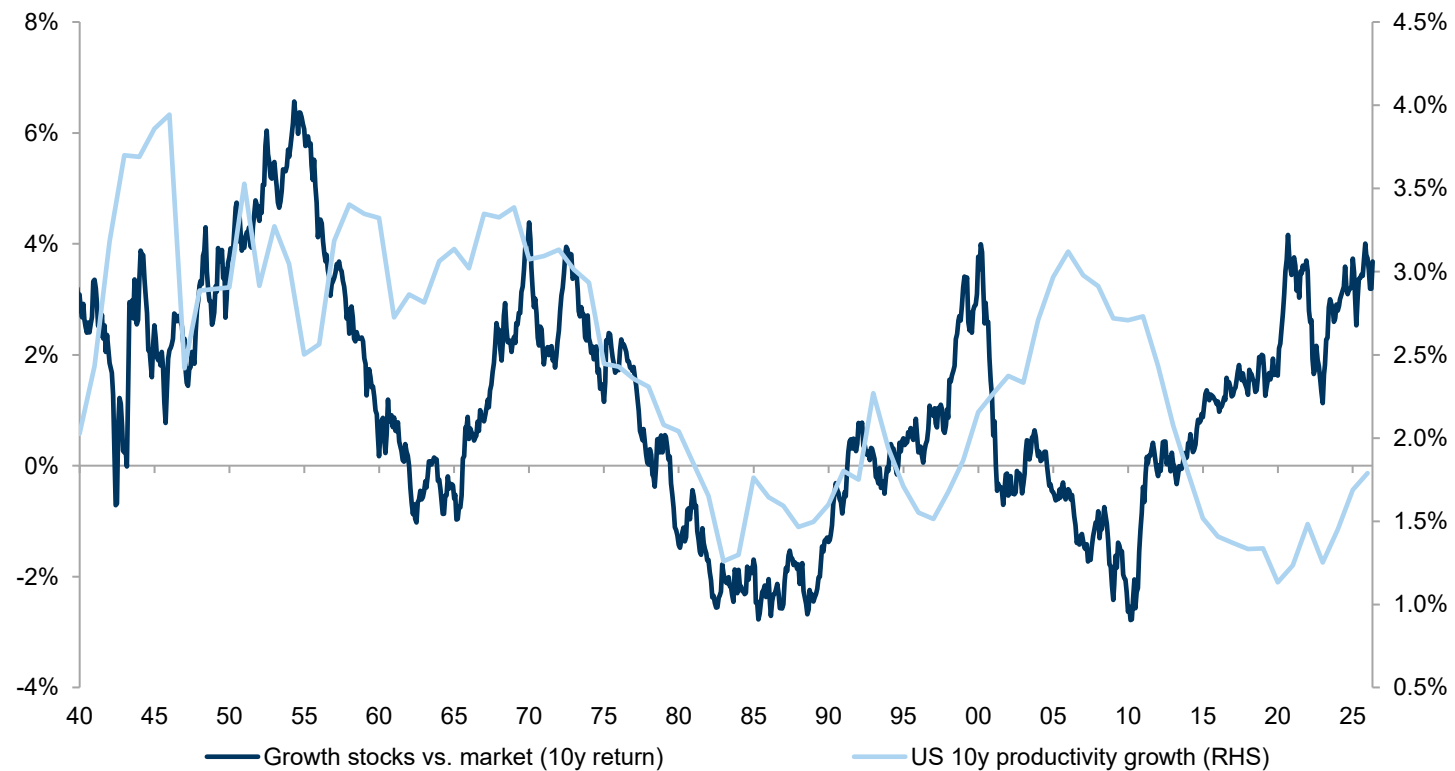
Relative return vs World Portfolio (risk-neutral scaling to match the World Portfolio vol, in brackets)



Source: Bloomberg, Datastream, Haver Analytics, Goldman Sachs Global Investment Research

Higher productivity growth tends to support growth stocks

Growth stocks often outperform around periods of strong productivity growth

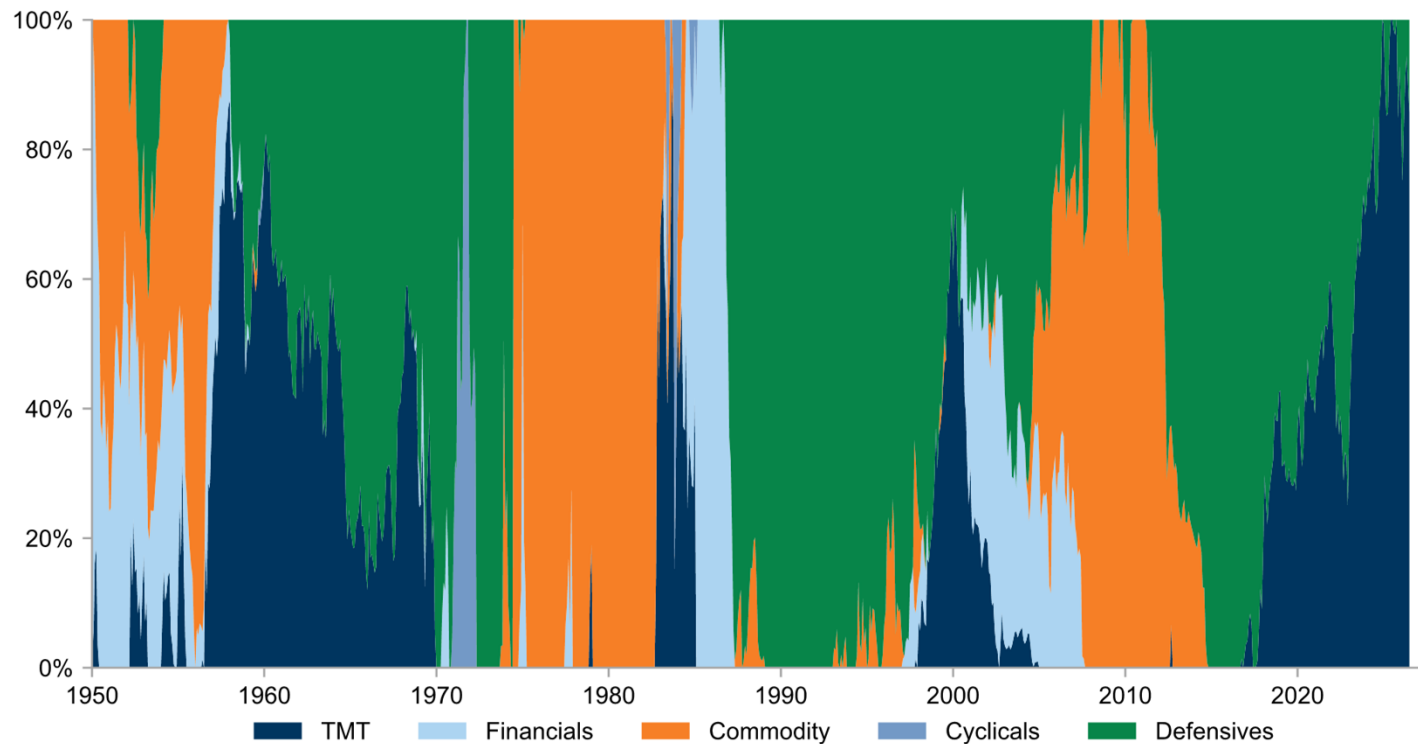


Note: MSCI US growth index. Before 1977 we use a value-weighted aggregate of the top quartile sub-sectors by dividend growth from Kenneth French.

Source: Kenneth French, US Bureau of Labor Statistics, Census Bureau, Haver Analytics, Datastream, Goldman Sachs Global Investment Research

Optimal sector weights varied even more over time

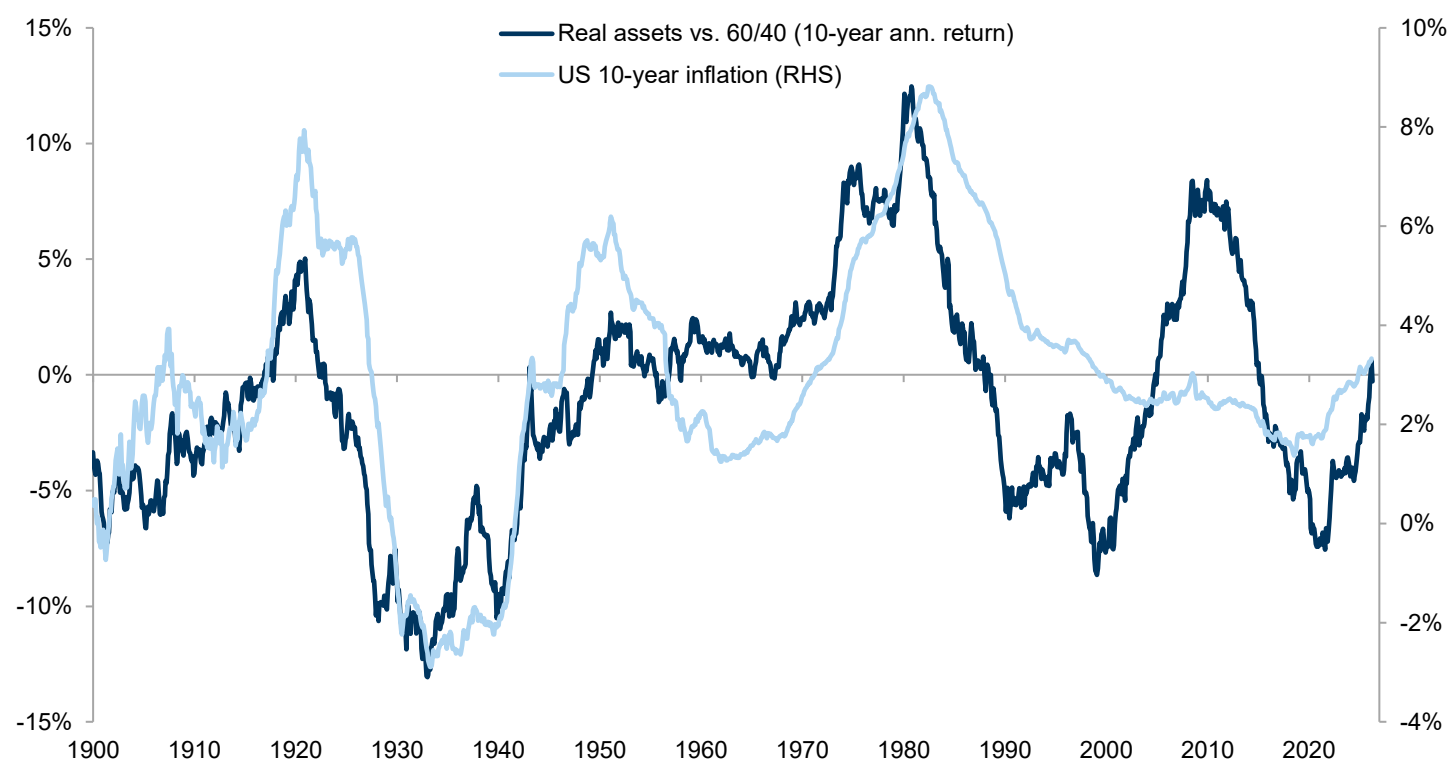
10-year rolling optimal sector weights for US equities



Source: Bloomberg, Datastream, Goldman Sachs Global Investment Research

With high inflation, real assets can offer uncorrelated real returns

Real assets have outperformed a 60/40 portfolio in periods of high and rising inflation



Note: 'Real assets' is an equal-weight average of S&P GSCI/ a broad commodities index (TR) since 1900, Gold since 1933, real estate since 1928, infrastructure/energy stocks since 1926, TIPS since 1959 using the a backcast; for details see Global Rates Notes: Introducing a backcast history of traded inflation.

Source: Haver Analytics, Kenneth French, Goldman Sachs Global Investment Research

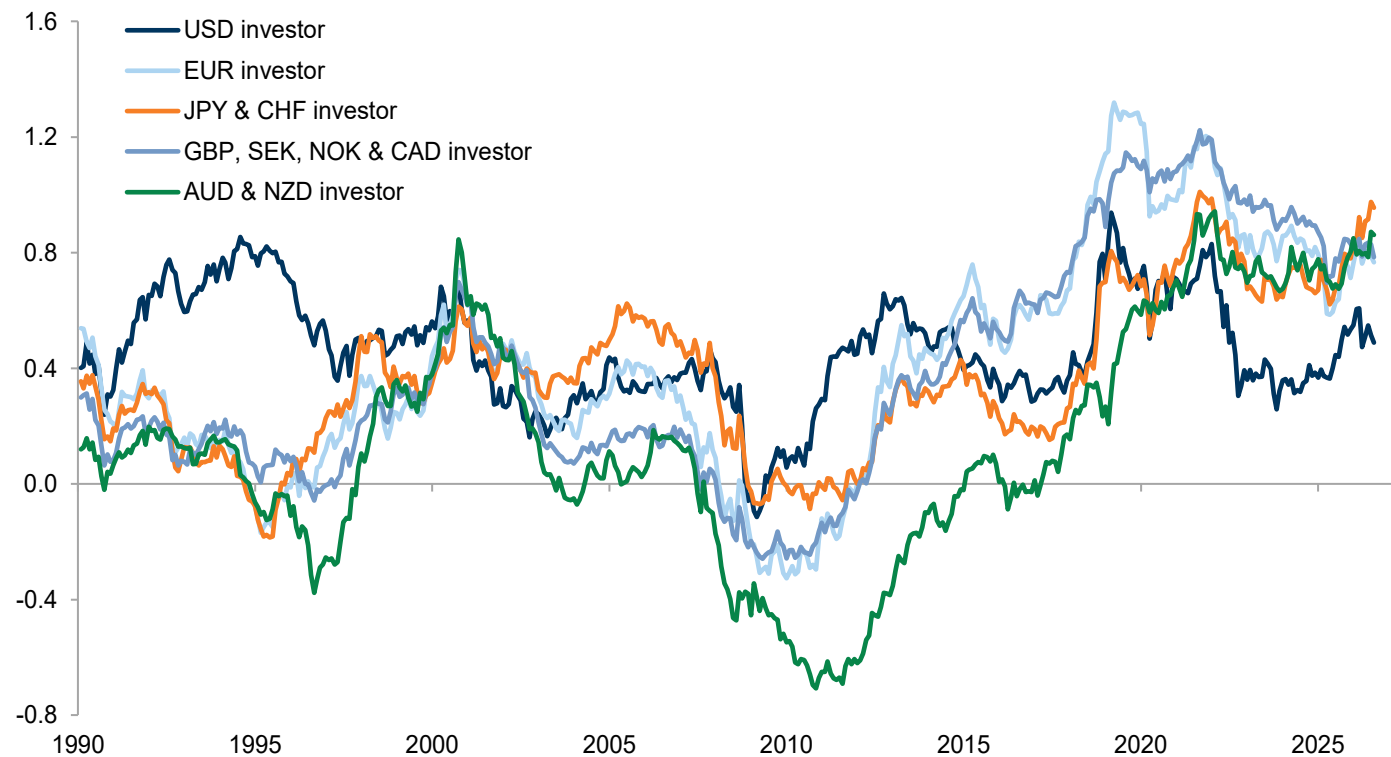
Benefits from broader diversification towards real assets

		World Portfolio	Average 10-year optimal weight adding:			
			Gold	S&P GSCI	Real Asset Stocks	Real Asset Index
Asset weights		<i>Current</i>				
US equity		36%	24%	24%	18%	21%
Non-US equity		20%	23%	24%	23%	18%
US bonds		19%	38%	37%	37%	36%
Non-US bonds		19%	5%	4%	6%	5%
Gold		6%	10%	-	-	-
S&P GSCI		-	-	10%	-	-
Real Asset Stocks		-	-	-	16%	-
Real Asset Index		-	-	-	-	20%
Performance metrics since 1950						
Annualised return		7.9%	8.7%	8.7%	9.2%	9.0%
Annualised volatility		8.2%	7.6%	7.8%	8.3%	7.8%
Tracking error			1.6%	2.7%	2.6%	2.4%
Sharpe ratio		0.45	0.58	0.57	0.59	0.61
Information ratio			0.44	0.27	0.47	0.41
5% CVaR of 1m returns		-5.0%	-4.5%	-4.8%	-5.1%	-4.7%
1y relative return	5th perc.		-1.9%	-4.1%	-3.5%	-3.5%
	95th perc.		4.1%	6.1%	6.8%	5.9%
10y relative return	5th perc.		-0.4%	-0.4%	0.0%	0.2%
	95th perc.		1.5%	2.1%	2.4%	1.8%
Hit ratio for real returns since 1950						
1y rolling real return	above 5%	50%	54%	55%	58%	56%
	below 0%	25%	22%	24%	22%	20%
10y rolling real return	above 5%	35%	42%	40%	55%	48%
	below 0%	5%	1%	2%	3%	1%

Source: Bloomberg, Datastream, Haver Analytics, World Gold Council, Goldman Sachs Global Investment Research

Risk-adjusted performance of the World Portfolio also varied materially based on investor domicile

Average of 10-year rolling Sharpe ratios of the World Portfolio

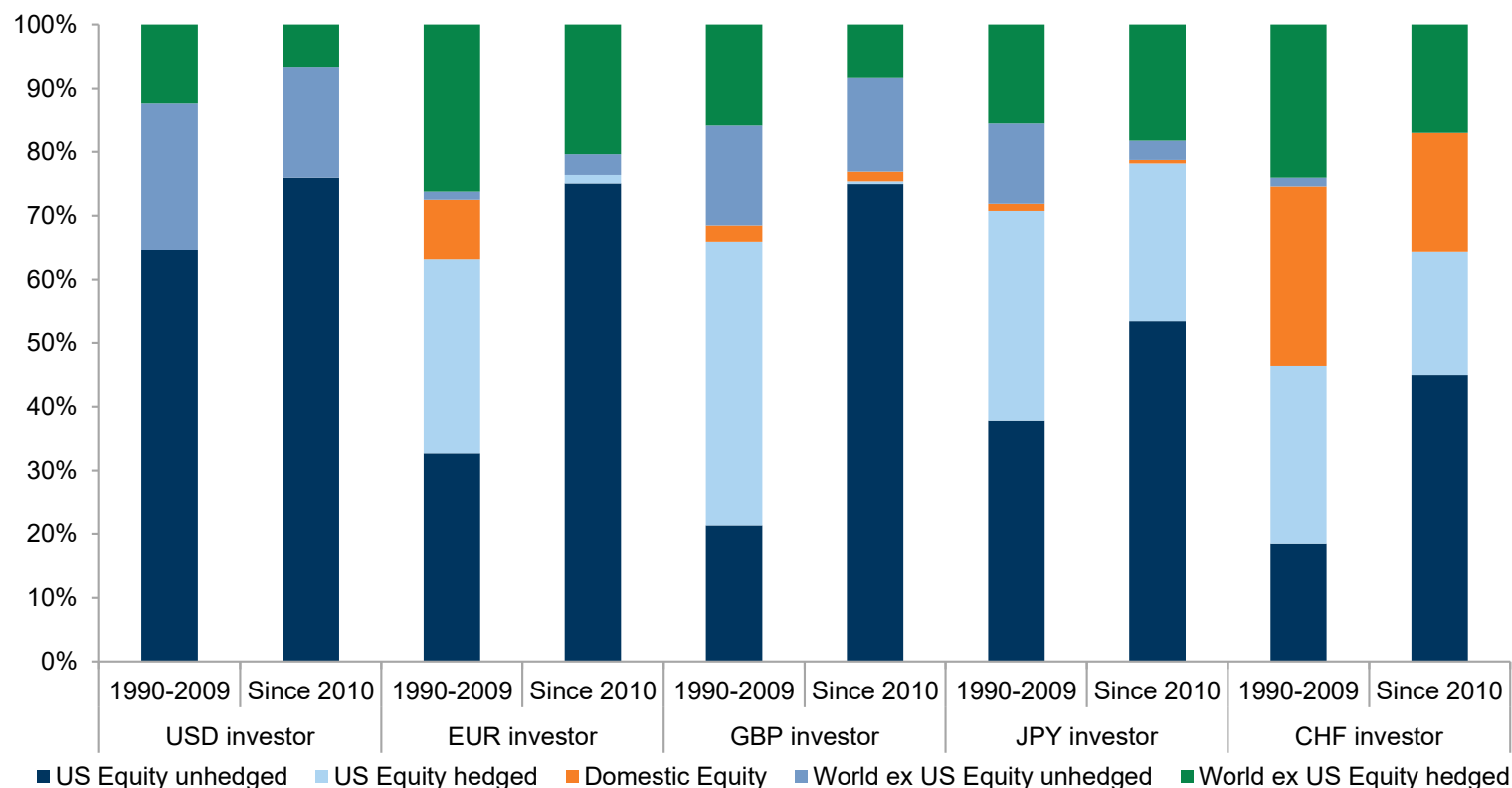


Source: Bloomberg, Datastream, Haver Analytics, Goldman Sachs Global Investment Research

Benefits from FX hedging vary over time and by investor domicile

FX risk can still impact optimal regional allocations

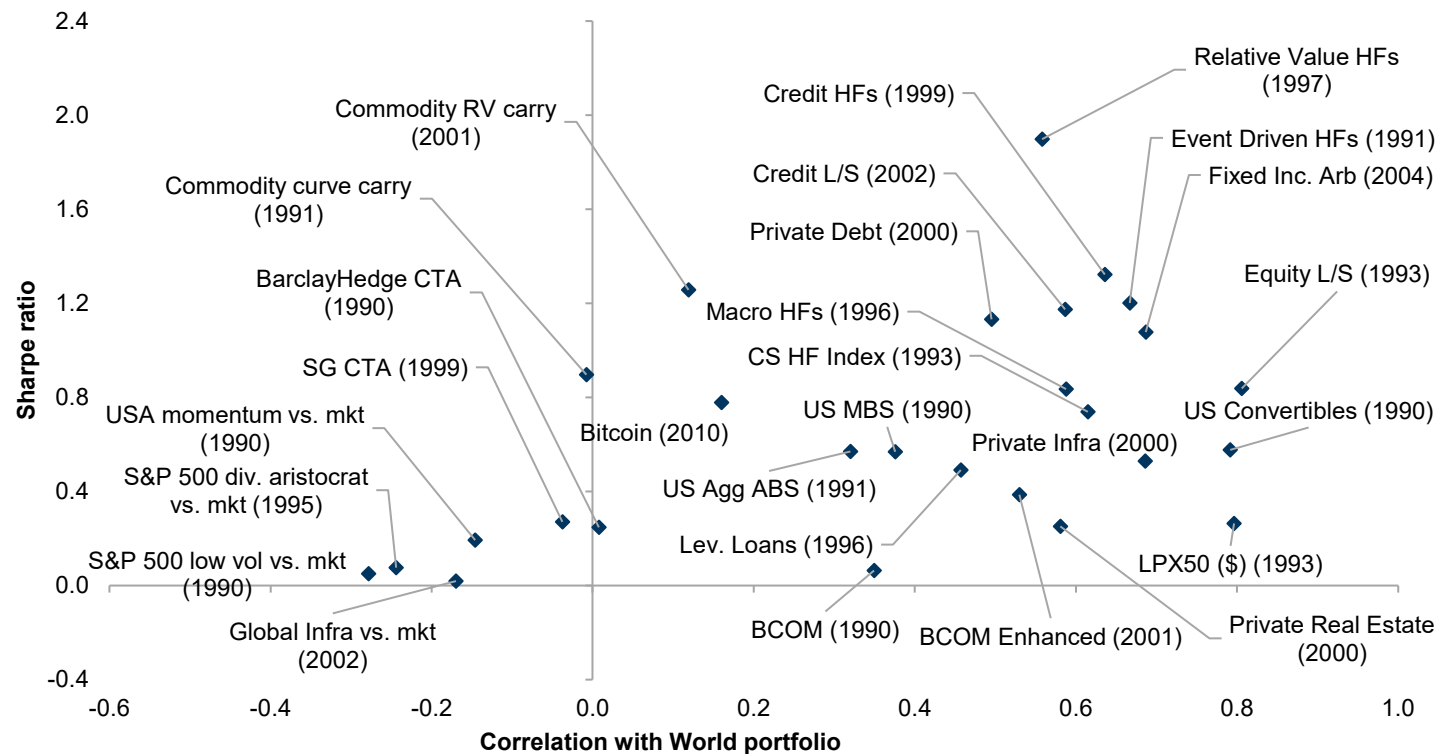
Average weights for optimal regional equity allocations: 1990–2009 versus 2010 to present



Source: Datastream, Goldman Sachs Global Investment Research

Assets and strategies outside the benchmarks might enhance the World Portfolio thanks to high Sharpe ratios or diversification benefits

Data since 1990 (where available, start date shown in brackets)

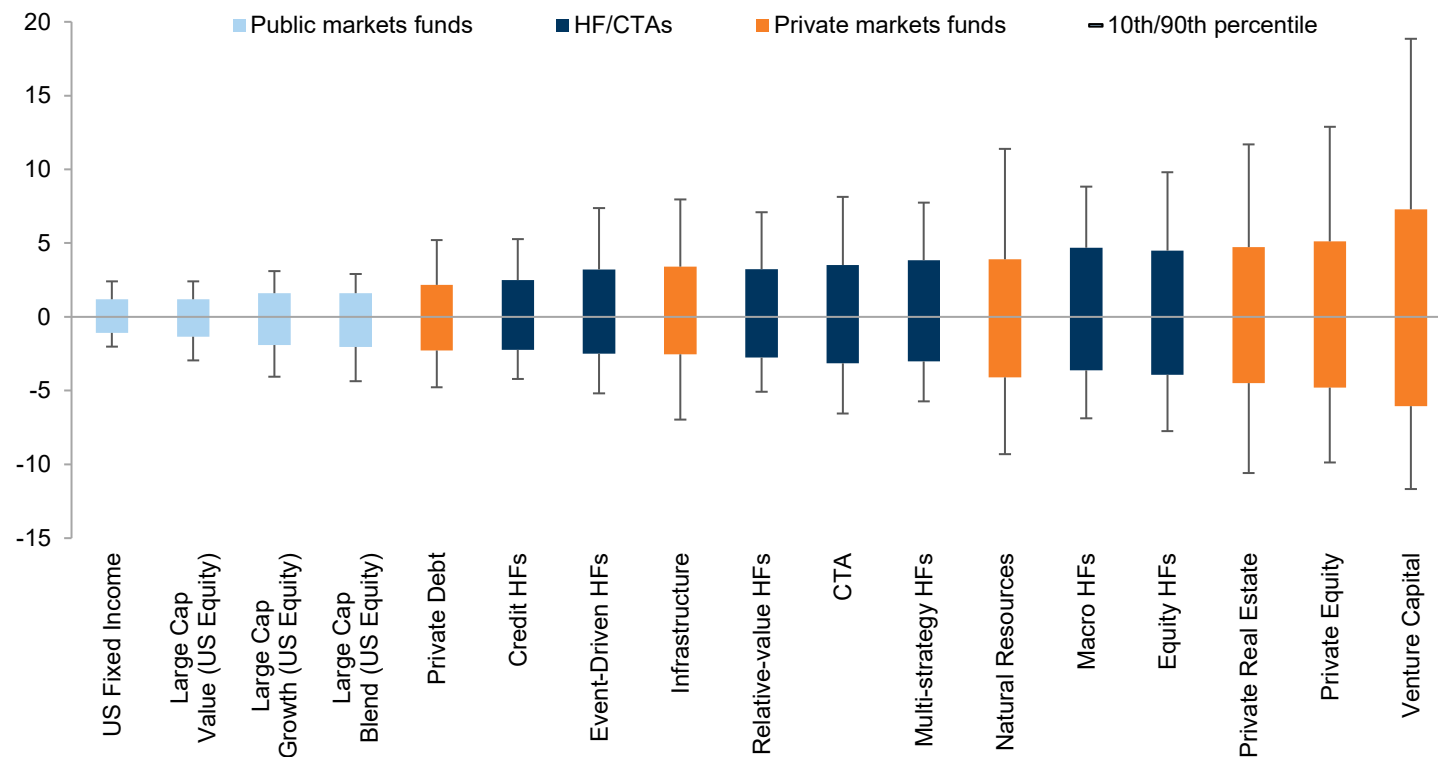


Note: *relative performance vs. the market, scaled by relative volatility, fully funded.

Source: Bloomberg, Datastream, Preqin, Goldman Sachs Global Investment Research

The performance of alternatives tends to be highly dispersed, especially for hedge fund strategies and private markets

Excess return vs. median (5-year annualised return/ net IRR, in %, since 2015)



Source: Bloomberg, Morningstar, Preqin, Goldman Sachs Global Investment Research

Cross-asset volatility hedges

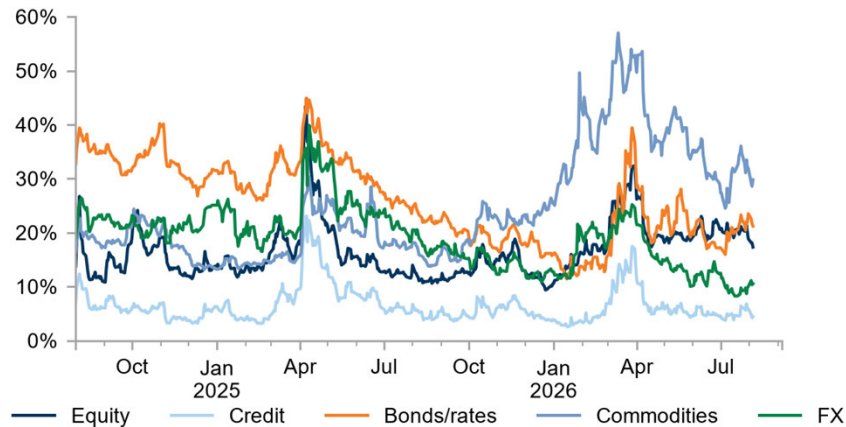
Cross-asset volatility overview

Cross-asset volatility overview

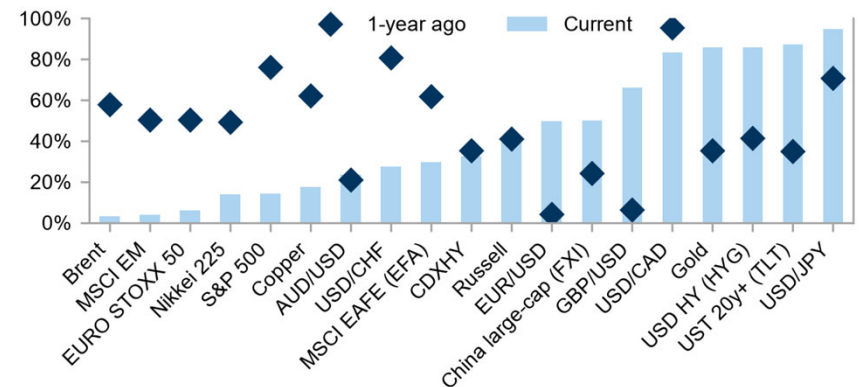
	Equities						Rates				Credit			Commodities			Currencies		
	S&P 500	EURO STOXX 50	Nikkei 225	FTSE 100	MSCI EM	China large-cap	USD 2y	USD 10y	EUR 2y	EUR 10y	CDX	CDXHY	iTraxx EU	WTI	Gold	Copper	EUR/USD	USD/JPY	GBP/USD
Implied (3-month ATM)																			
Current:	13.7	14.2	36.3	9.6	34.0	17.2	4.4	3.8	4.1	3.3	0.8	4.1	1.0	65.0	24.1	16.4	4.9	10.9	6.2
Percentile:	61%	55%	96%	38%	99%	23%	58%	36%	78%	61%	17%	18%	28%	95%	92%	37%	23%	82%	26%
1M change:	-3.6	1.1	-1.4	-1.3	1.5	-1.5	-0.2	-0.1	1.5	0.7	0.1	0.2	0.5	27.4	-5.9	-1.3	-1.0	6.1	-0.3
Average:	14.6	15.3	18.5	12.5	14.5	24.6	4.2	4.8	2.5	3.5	1.7	8.3	1.7	37.0	14.8	19.3	6.5	8.0	7.9
95th:	28.8	30.1	33.7	24.8	27.4	44.4	8.9	8.3	7.3	7.4	3.7	19.1	3.7	71.8	29.0	31.5	10.9	14.4	12.9
5th:	5.8	7.6	9.4	6.7	8.0	13.3	0.7	2.6	0.4	1.6	0.6	3.0	0.6	18.7	8.2	11.7	3.6	4.0	4.9
Realised (1-month)																			
Current:	13.7	14.2	36.3	9.6	34.0	17.2	4.4	3.8	4.1	3.3	0.8	4.1	1.0	65.0	24.1	16.4	4.9	10.9	6.2
Percentile:	61%	55%	96%	38%	99%	23%	58%	36%	78%	61%	17%	18%	28%	95%	92%	37%	23%	82%	26%
Average:	14.6	15.3	18.5	12.5	14.5	24.6	4.2	4.8	2.5	3.5	1.7	8.3	1.7	37.0	14.8	19.3	6.5	8.0	7.9

As of 04-Aug-2026

Aggregate cross-asset volatility, 0%/100% = min/max since 2010



3m 25-delta normalised put/call skew. 5y percentile

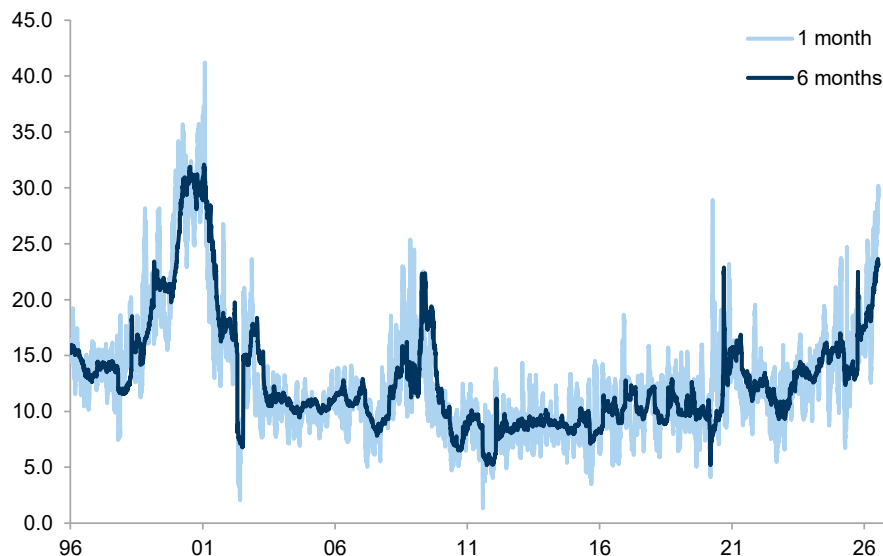


Source: Bloomberg, Datastream, Goldman Sachs FICC and Equities, Goldman Sachs Global Investment Research

Equity dispersion has been high, pushing implied correlations to all-time lows

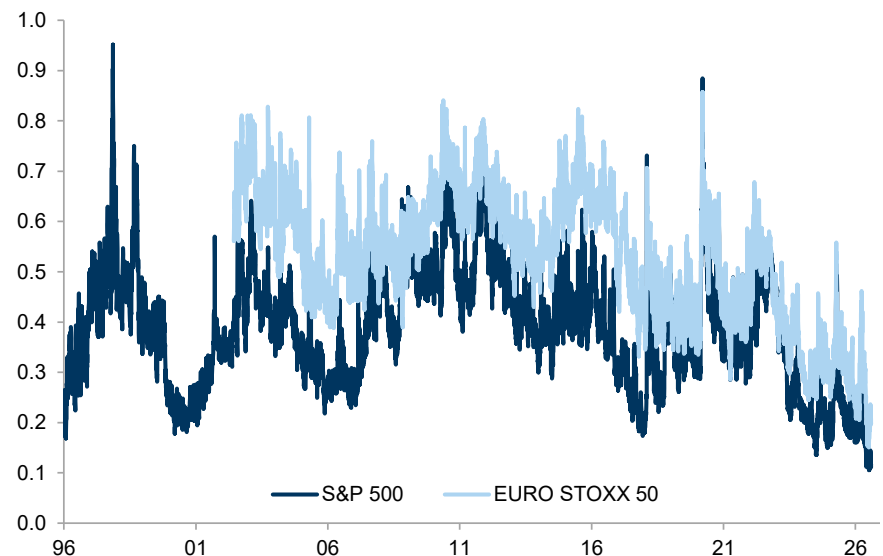
Realised dispersion is at the highest level since the early 2000s

Average realized vol of underlying stocks – realised vol of the S&P 500



Implied correlations are at all-time lows

3-month implied correlation at the 50-delta call

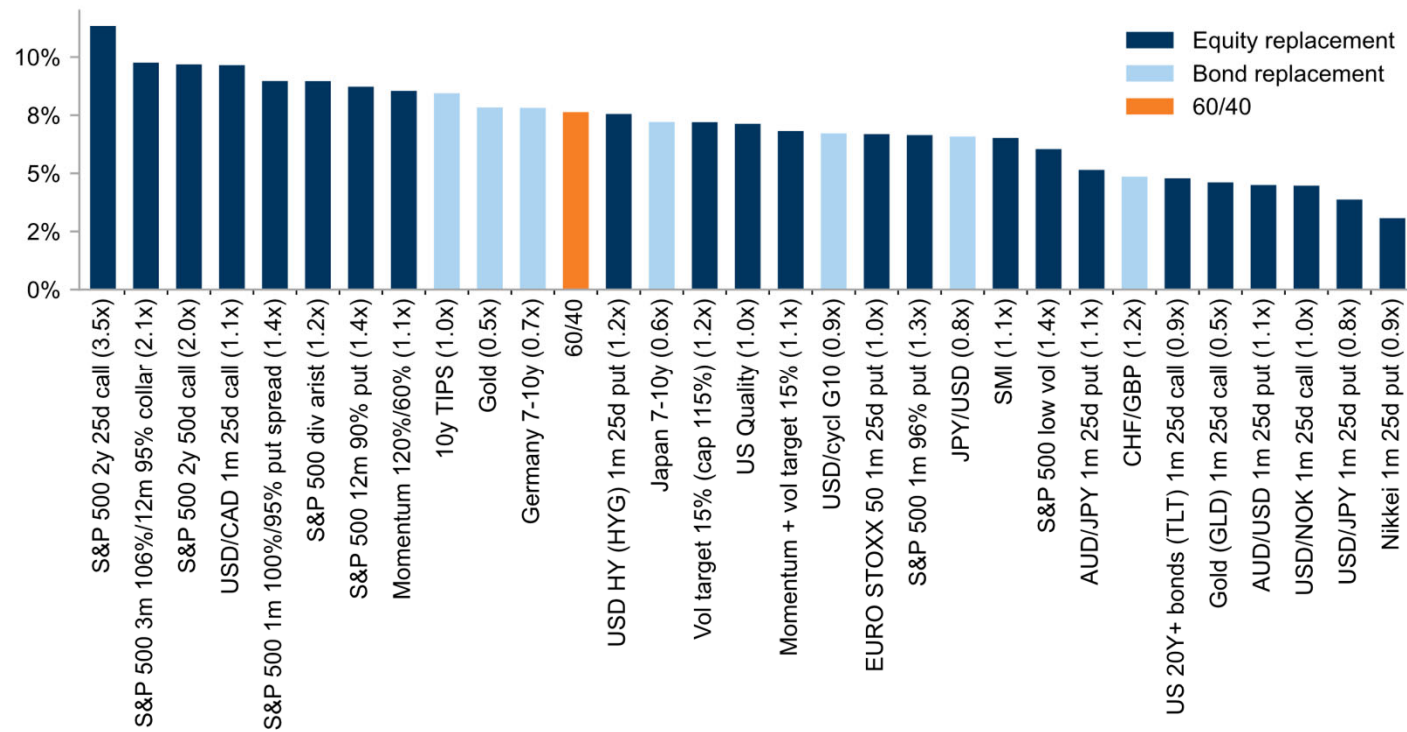


Source: Bloomberg, Goldman Sachs Global Investment Research

Few risk mitigation strategies have worked YTD

YTD performance of a US 60/40 portfolio with equity or bonds replaced by risk-mitigating strategies

Equity/bond replacements leveraged to match the 5-year vol of S&P 500/US 10y bonds (latest scale in brackets)

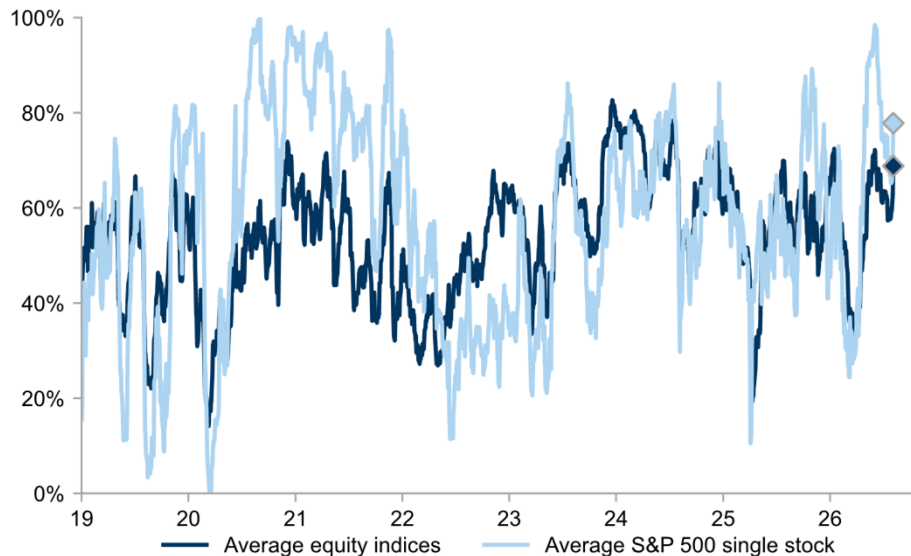


Source: Haver Analytics, Bloomberg, Goldman Sachs Global Investment Research

Put spread collars attractive to protect after strong rally

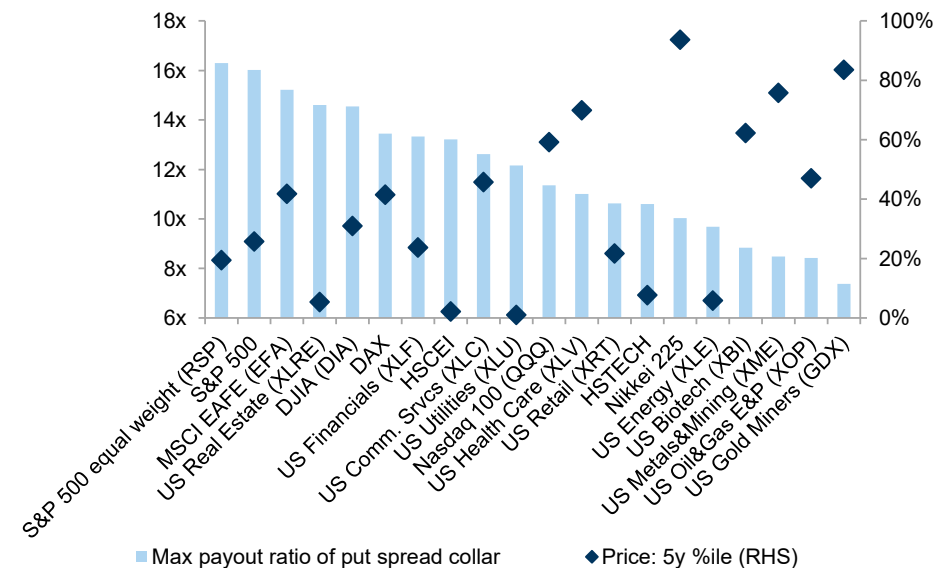
Call option positioning has turned sharply bullish since March, in particular for single stocks

Average 5y percentile of Spot/Vol correlation, Call/Put ratio and Call skew across 60 ETF and equity indices



Selective put spread collars look attractive in the current environment

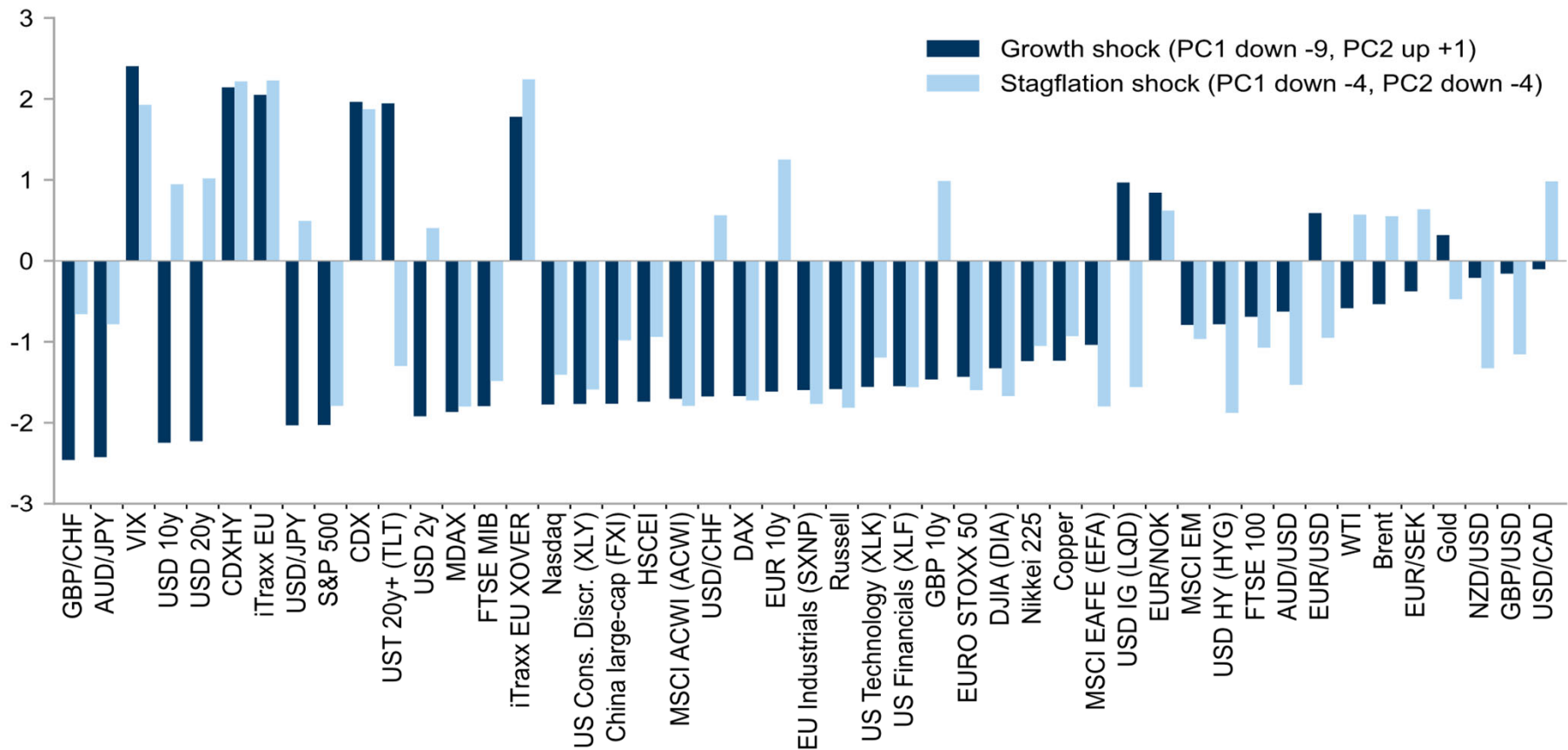
3m 40D/15D put spread + short 3m 25D call for the 20 markets with the highest Call option positioning



Source: Bloomberg, Goldman Sachs FICC and Equities, Goldman Sachs Global Investment Research

Cross-asset hedges for a growth or stagflation shock in the summer

Cross-asset move implied by the beta to PC1 and PC2 / current ATM 3m implied vol



Source: Goldman Sachs Global Investment Research

Key views and forecasts across assets

Overview of key GS views across assets

	Allocation Weighting		Current Level	Forecasts			Forecast Upside/Downside (%)		
	3m	12m		3m	6m	12 m	3m	6m	12 m
Equities	N	OW			Index level			Total Return	
S&P 500 (\$)	OW	OW	7710	7600	8000	8300	-1.1	4.3	8.8
Stoxx Europe 600 (€)	UW	UW	658	640	645	660	-2.0	-0.4	3.5
MSCI Asia-Pacific ex. JP (\$)	OW	OW	860	980	1030	1080	13.9	19.6	25.9
Topix (¥)	OW	OW	4056	4200	4300	4500	4.1	7.1	13.2
10 Year Government Bonds	N	N			Yield (%)			Total Return	
US	OW	OW	4.67	4.43	4.38	4.28	3.2	4.8	8.2
Germany	OW	N	3.13	2.97	3.00	3.00	2.3	2.9	4.8
Japan	UW	UW	2.77	2.50	2.48	2.36	3.4	4.6	7.8
UK	N	OW	4.94	4.56	4.50	4.40	4.4	6.3	10.1
Corporate Bonds	N	UW			Spread			Total Return	
Bloomberg USD IG	OW	OW	77	84	81	78	2.6	4.5	8.4
Bloomberg USD HY	OW	OW	263	301	293	280	1.1	3.0	6.9
Bloomberg EUR IG	UW	UW	78	90	87	84	1.0	2.0	4.1
Bloomberg EUR HY	UW	UW	286	318	313	305	0.6	1.9	4.6
EM Hard Currency Sovereign	N	N	236			260	2.8	4.4	7.7
Commodities	N	N			Spot price			Spot Return	
WTI (\$/bbl)	N	N	79	76	73	68	-3.7	-7.5	-13.8
Brent (\$/bbl)	N	N	83	80	78	73	-3.1	-5.5	-11.6
Copper (\$/mt)	N	N	14,227	13,700	13,800	13,800	-3.7	-3.0	-3.0
Gold (\$/troy oz)	N	OW	4,264	4,840	5,005	5,210	13.5	17.4	22.2
FX	-	-			Spot rate			Spot Return	
EUR/USD	-	-	1.15	1.14	1.12	1.12	-1.2	-2.9	-2.9
USD/JPY	-	-	158.2	162	163	165	2.4	3.0	4.3
GBP/USD	-	-	1.35	1.33	1.29	1.28	-1.2	-4.2	-4.9
AUD/USD	-	-	0.70	0.72	0.73	0.74	2.3	3.7	5.1
USD/BRL	-	-	5.09	4.90	5.00	5.00	-3.8	-1.9	-1.9
USD/INR	-	-	95.2	94.0	95.0	96.0	-1.3	-0.2	0.8
USD/CNY	-	-	6.75	6.80	6.70	6.50	0.7	-0.8	-3.7
Cash	N	N			Spot rate			Total Return	
US 3-month Tbill	-	-	-	-	-	-	0.96	1.98	4.06
Germany 3-month Bubills	-	-	-	-	-	-	0.59	1.23	2.56

Pricing as of August 6th, 2026

Source: Bloomberg, Datastream, Bloomberg-Barclays, ICE-BAML, iBoxx, Goldman Sachs Global Investment Research.

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August 07, 2026

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